ARPES viewer — User Guide

中文版 / Chinese version · *Version française*

Last updated: 25 October 2026 — for ARPES viewer v1.0.0

The version and its date are in the main window's title bar and at the bottom of the main window. Bug reports, questions and the latest version: https://github.com/DingPei1995/ARPES-data-browser (please give the version number when reporting a bug).

This guide walks through the program's main features with screenshots, in the order you would normally use them: loading data, looking at it, converting it to momentum, analysing it, and making figures. The technical background for each step (formulas, test results, design choices) is in the README and CHANGELOG.

Every screenshot was taken from the running program, using the synthetic data written by `devtools/demo_data.py`, so you can follow along without a beamtime (see Trying it with demo data).

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1. Installing and starting

You need Python 3.9 or newer and five packages:

```
pip install PyQt5 pyqtgraph numpy h5py scipy
```

Then, from the `ARPES_viewer` folder:

```
python ARPES_viewer.py
```

The program reads:

Beamline / source	Files
SOLEIL **ANTARES**	NeXus/HDF5 `*.nxs` (cuts, deflector maps, spatial scans)

Beamline / source	Files
SOLEIL **CASSIOPEE**	Scienta SES text files — single spectra, and numbered folders assembled into a map or a photon-energy scan
CASSIOPEE **spin end station** (MBS A-1)	`*.krx` and `*.txt` — cuts, deflector maps and spin-resolved EDCs
This program	its own saved `*.nxs` format

2. Trying it with demo data

```
python devtools/demo_data.py           # writes ./demo_data/*.nxs
```

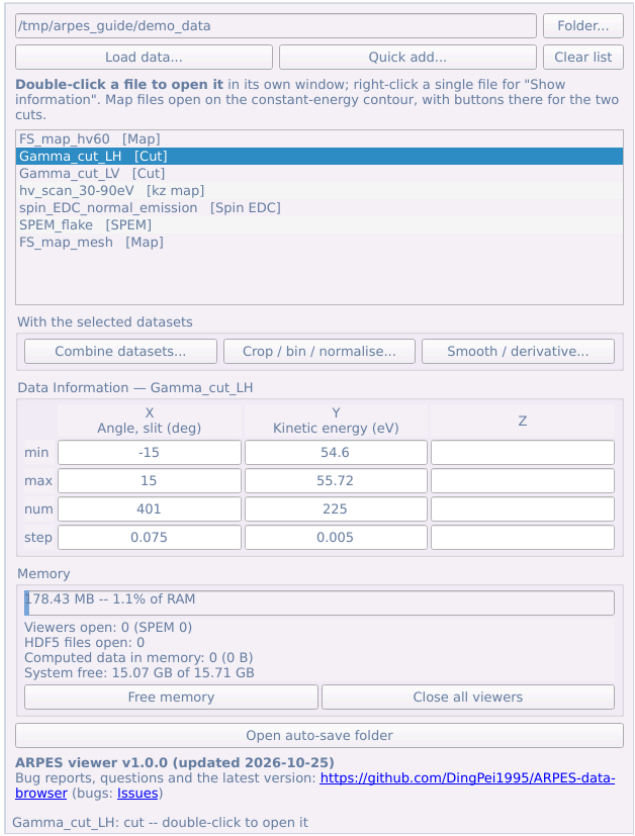
This writes six files of synthetic data: a Fermi-surface map, a pair of cuts taken with LH and LV polarisation, a photon-energy scan, a spin-resolved EDC, a small real-space scan and a map with a detector mesh printed on it. They come from a toy band structure (a hexagonal electron pocket and a hole band at Γ, hole pockets at K) with a Fermi cut-off, the analyser resolution and Poisson noise, and open exactly like measured data.

To re-take every screenshot in this guide after changing a window:

```
python devtools/guide_screenshots.py    # writes docs/images/*.png
```

3. The main window

The main window is a **browser**: one row per *dataset*, not per file (one `.nxs` file often holds several measurements, and each gets its own row). The kind is written after the name: `[Cut]`, `[Map]`, `[kz map]`, `[k-map]`, `[SPEM]`, `[EDC]`, `[Spin EDC]` …



Action	How
Open a dataset in its own window	**double-click** the row (or right-click → **Open**)
Stack several cuts into one map (temperature, gate voltage, photon energy → kz map ...), or join a map / kz map measured in segments	select them → **Combine datasets...** (or right-click → **Combine N datasets...**)
Everything else that works on datasets	**right-click** the row (or a selection of rows)
Read the axes: first/last value, points, step	right-click → **Show information** (fills *Data Information*)
Change the grid or scale — crop to a range, bin points, normalise each spectrum to itself — for several datasets at once	**Crop / bin / normalise...**
Bring out the bands — smooth, second derivative, curvature, background subtraction, symmetrise	**Smooth / derivative...**
See and free memory	the **Memory** box at the bottom
See which version you are running, report a bug, get the latest version	the title bar and the last line of the window (link to GitHub)

The three buttons under the list sit in a box called **With the selected datasets**: each acts on the rows selected above it, and each makes new rows — the originals are never changed.

Which name does what

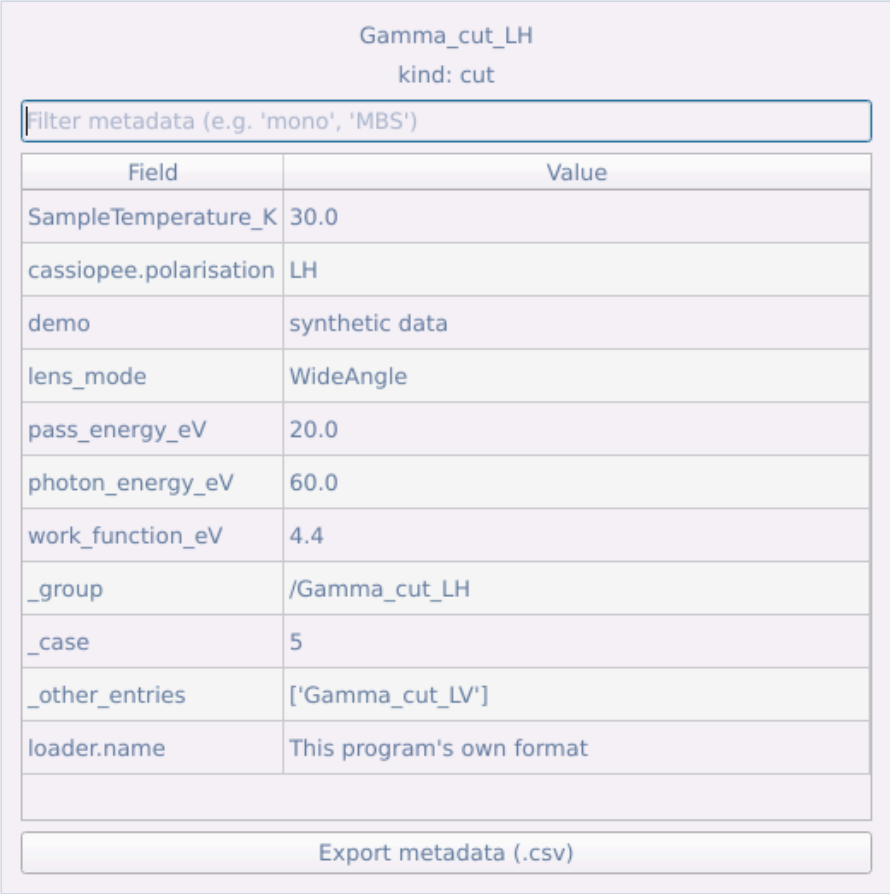
Several buttons were renamed (v1.0.0) so that the name says what happens. If you knew the old names:

Now	Was	What it does
Combine datasets...	Combine data...	stack cuts into a map, or join pieces of one measurement
Crop / bin / normalise...	Data operations...	changes the grid or scale: crop (truncate), bin (compress), self-normalise
Smooth / derivative...	Process...	filters that bring out bands: smooth, derivative, curvature, background, symmetrise
Functions section **Transform**	Data operations	conversions and corrections in a viewer's Functions menu
Fermi edge fit...	Fermi level...	fit E_F (and resolution) on a cut
Convert to momentum (k)...	Cut / Map k conversion...	angle → k∥
Align Fermi levels (kz map)...	kz map processing...	fit and align E_F of every photon energy
Convert to momentum (kz, k∥)...	kz -> momentum...	photon-energy scan → (k_z, k∥)
Straighten bands (FS correction)...	FS correction...	flatten a curved Fermi edge or feature
Remove detector grid...	De-grid... / De-grid map...	remove the MCP / mesh pattern

Now	Was	What it does
Multi-slice figure...	Slice figure...	a page of slices through a map
Freeze slice in a new window	Open slice in a new panel	a frozen copy of the slice on screen
Plot tools (viewer top row)	— (new)	guide lines and labels drawn on the live plot
Figure composer... (frozen slice), **Open in figure composer...** (list)	Plot tools, Plot as a figure...	make a figure for a paper
Figure tools... (inside the figure composer)	Plot tools	size, fonts, ticks, colour bars, annotations of that figure

The right-click menu only offers what fits the selection. A greyed entry says why when you hover over it — for example, *Cut arithmetic* needs exactly two cuts, the *3D view* needs a map.

Show information also opens the full metadata of the dataset in a window of its own, with a filter box and **Export metadata (.csv)**:



Clear list asks first, and offers to save anything computed in this session that has not been saved yet. Files on disk are never deleted.

Combining datasets: stacking and joining

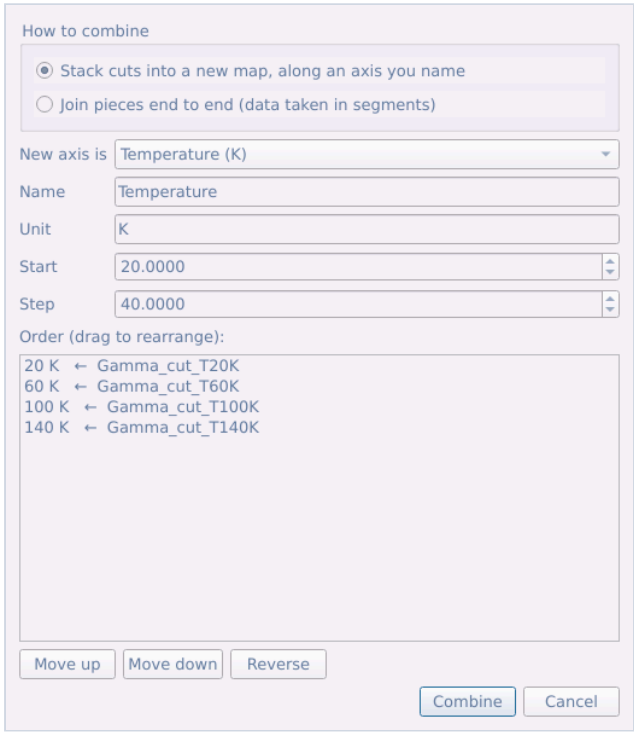
Combine datasets... (under the list, or right-click → **Combine N datasets...**) does two different things, and asks which:

- **Stack** — *cuts only*. Each cut becomes one slice of a new map, along an axis you name: a temperature series, a gate-voltage series, one cut per photon energy. The result is looked at like any map — sliced at constant energy, or along the new axis.
- **Join** — *cuts, maps, kz maps or k-maps, all of one kind*. The pieces of one measurement that was taken in segments are put back end to end along one of their own axes: a map whose deflector range was scanned in two parts, a kz map whose photon-energy range was split over several runs, a cut whose energy window was taken in two halves.

Stacking cuts

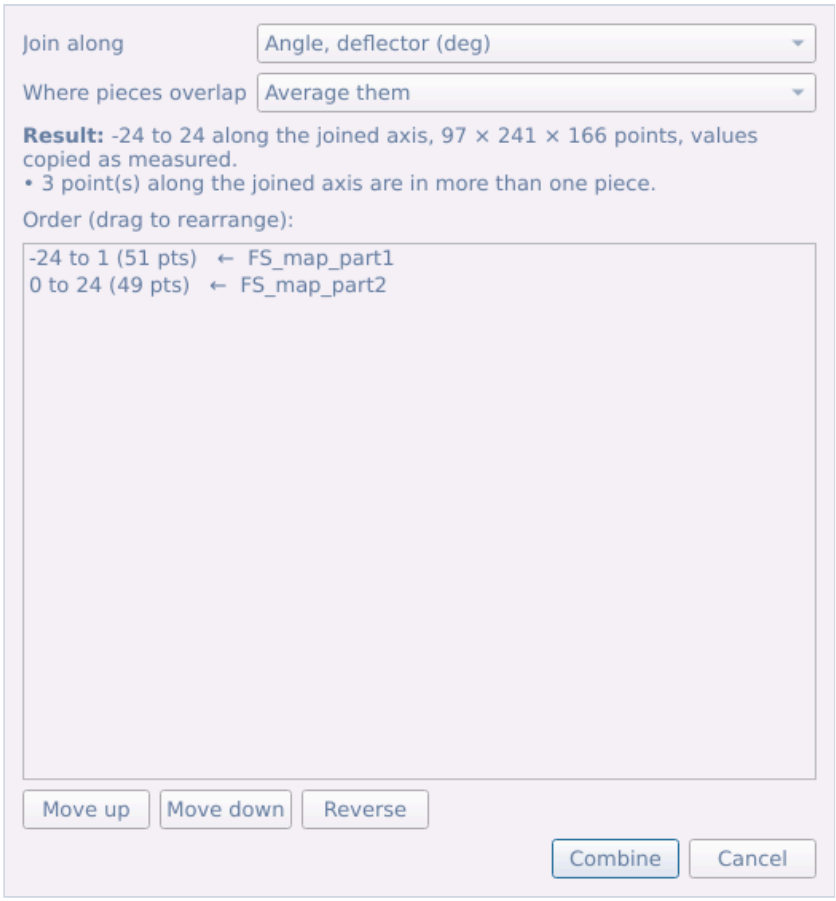
1. Select the cuts in the list (Ctrl/Shift-click) and press **Combine datasets...**. Stacking needs the same axes point for point (same number of points, same values, same units); if they differ, *Stack* is greyed with the first difference named, and *Join* is offered instead.
2. It asks whether to combine them **in the list's order**. *Yes* keeps that order; *No* lets you choose it.
3. Choose **Stack**, then set the new axis: **New axis is** (temperature, gate voltage, photon energy, delay, position, other — this fills in *Name* and *Unit*, which you can change), **Start** and **Step**. Each cut is listed with the value it gets. If you chose to reorder, **drag** the rows (or use **Move up** / **Move down** / **Reverse**). Press **Combine**.

A photon-energy series of angle cuts becomes a `kz map`, so *Align Fermi levels (kz map)* and *Convert to momentum (kz, k∥)* take it (see §13); any other axis gives a `map` whose first axis is marked as not an angle, so no k conversion is offered for it.



Joining pieces of a measurement

1. Select the pieces (all maps, or all kz maps, or all cuts …) and press **Combine datasets...**. Maps and kz maps go straight to *Join*.
2. **Join along**: the axis the measurement was split along — by default a map's scanned (deflector) axis or a kz map's photon energy; for cuts, choose the angle or the energy axis.
3. **Where pieces overlap**: *Average them*, or keep the earlier / later piece in the order shown (drag the rows to change it).
4. The dialog lists each piece's range and shows the **Result** — the joined range and the number of points — with notes on anything it had to do. Press **Combine**.



How the axes are handled:

- Pieces on one common grid (same step, ranges that meet or overlap by whole points — the usual case) are **copied as measured**, nothing interpolated.
- A piece with a different step or a half-step offset is **interpolated** onto the common axis (the first piece's step), and the dialog says so.
- If the **other axes** differ — an energy window a few meV different between runs, a different number of slit channels — every piece is resampled onto the range they **all** cover, and the dialog says so.
- A stretch that no piece covers is a **gap**, left at zero and reported.
- A joined kz map is still a `kz map`: run *Align Fermi levels (kz map)* on the joined result rather than on the pieces (per-photon-energy fit results of the pieces are not carried over).

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4. Loading data

Load data... opens the loader. **Quick add...** skips it and adds the chosen files with everything detected automatically.



1. **Files** — pick one or many (*Select file(s)...*). The file filter is built from what the installed readers can open.
2. **Reader** — detected automatically, and the result is shown (*Detected: 3 × This program's own format*). Override it only if a file is being misread.
3. **Datasets found** — what is inside each file, read from its *structure* only. This is instant even for a multi-GB scan; nothing is read until you open a row.
4. **Axes** — *Axis order* reorders the array's dimensions once, at the door. *First axis of a map is* says what was scanned: a deflector angle (the default), a **photon energy**, a temperature, a gate voltage, a delay… This keeps the right unit everywhere afterwards, and stops a k conversion being offered for an axis that is not an angle.

CASSIOPEE folders. Select any file of a numbered series (`name_1_ROI1_.txt`, `name_2_ROI1_.txt`, …) and the whole folder is listed once, as one assembled cube. Whether it becomes an angle map or a kz map is read from the `_i` parameter files next to the spectra. The row is named with the photon energy and polarisation read from those files — `FineMap 117eV LH`, or `KzScan 40-120eV LH` for a photon-energy scan — so repeats of the same scan at another energy or polarisation can be told apart.

ANTARES photon-energy (PhD) files. A PhD `.nxs` file (one `Phd_NNNN` entry per photon energy) is listed as one `kz map`: the frames are sorted by photon energy and each kinetic-energy axis is referred to E_F with its own hν (work function 4.5 eV, which the k_z conversion then uses). Each frame still opens on its own as a cut.

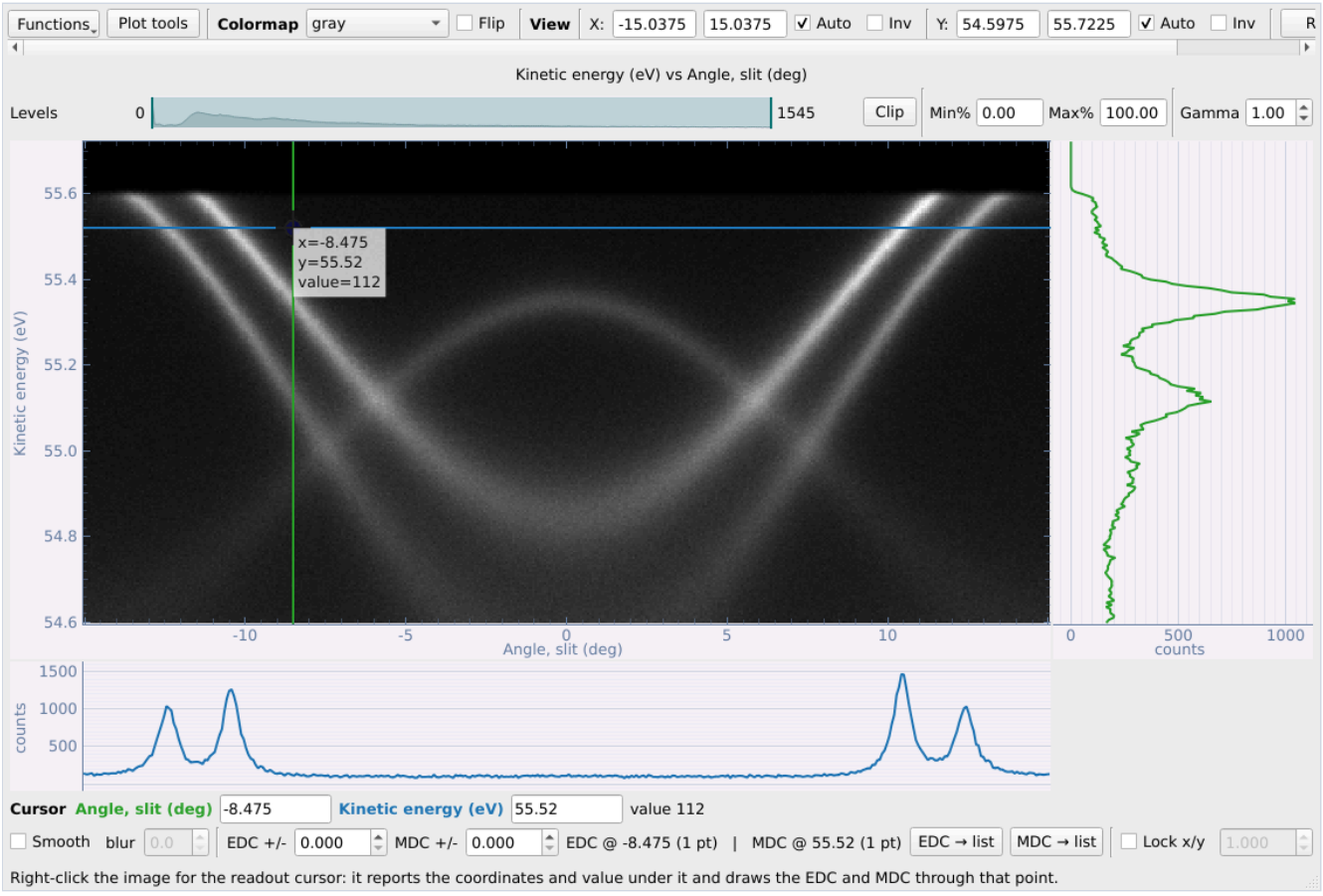
Data kinds

Kind	Axes	What it is
`cut`	angle (or k) × E	one spectrum
`map`	angle × angle × E	a deflector or polar-angle scan

Kind	Axes	What it is
`k_map`	kx × ky × E	a map converted to momentum
`kz_map`	hν × angle × E	a photon-energy scan
`kz_map_k`	k_z × k∥ × E	that scan converted to momentum
`spem_4d`, `spem_1d`	x (× y) × angle × E	real-space scans
`edc`, `mdc`, `spin_edc`	E (or k) × channels	curves, with optional σ channels

5. What the viewers have in common

Each dataset opens in its own window, so you can put a map, its cuts and another file side by side.



- **The top row**: the **Functions** menu (everything the window can do), **Plot tools** (guide lines and labels on this plot — see below), the colormap (per window) with **Flip**, and the view controls — axis ranges, **Auto**, **Inv** (invert), **Reset**, **Grid**.
- **Above each image**: the **Levels** bar (drag the two handles, or type **Min%/Max%** percentiles; **Clip** applies them) and **Gamma**. A min/max you set is kept while you move the slice (the energy of a contour, the angle of a cut, the cursor of a spatial map), so every position is on the same colour scale; drag the handles back to both ends, or set 0%–100%, to follow each frame's own range again.
- **Under each image**: **Smooth** (display only, does not change the data) and **Lock x/y** (aspect ratio).

- **Readout cursor** — right-click the image → *Readout cursor*. It reports the value under it and, on a cut, draws the **EDC** (right) and **MDC** (below) through that point. **EDC ±** / **MDC ±** integrate over a window, shown as a shaded band on the image. Type a position into the **Cursor** row and press Enter to put it exactly there.
- **EDC → list** / **MDC → list** put the curve on screen in the main list as a dataset of its own (see §15).
- **Right-click the image → Export** saves what the panel shows — data (`.nxs`), an image, or vector axes (`.eps`); see §17.
- **Functions → Slice → Save slice to the main list** / **Freeze slice in a new window** keeps a copy of the slice on screen. The frozen window (and *Open this position in a new window* on a spatial map) opens with the panel's min/max and gamma, so the two can be compared directly. It has its own **Plot tools**, and a **Figure composer...** button to make a figure of it (§17).

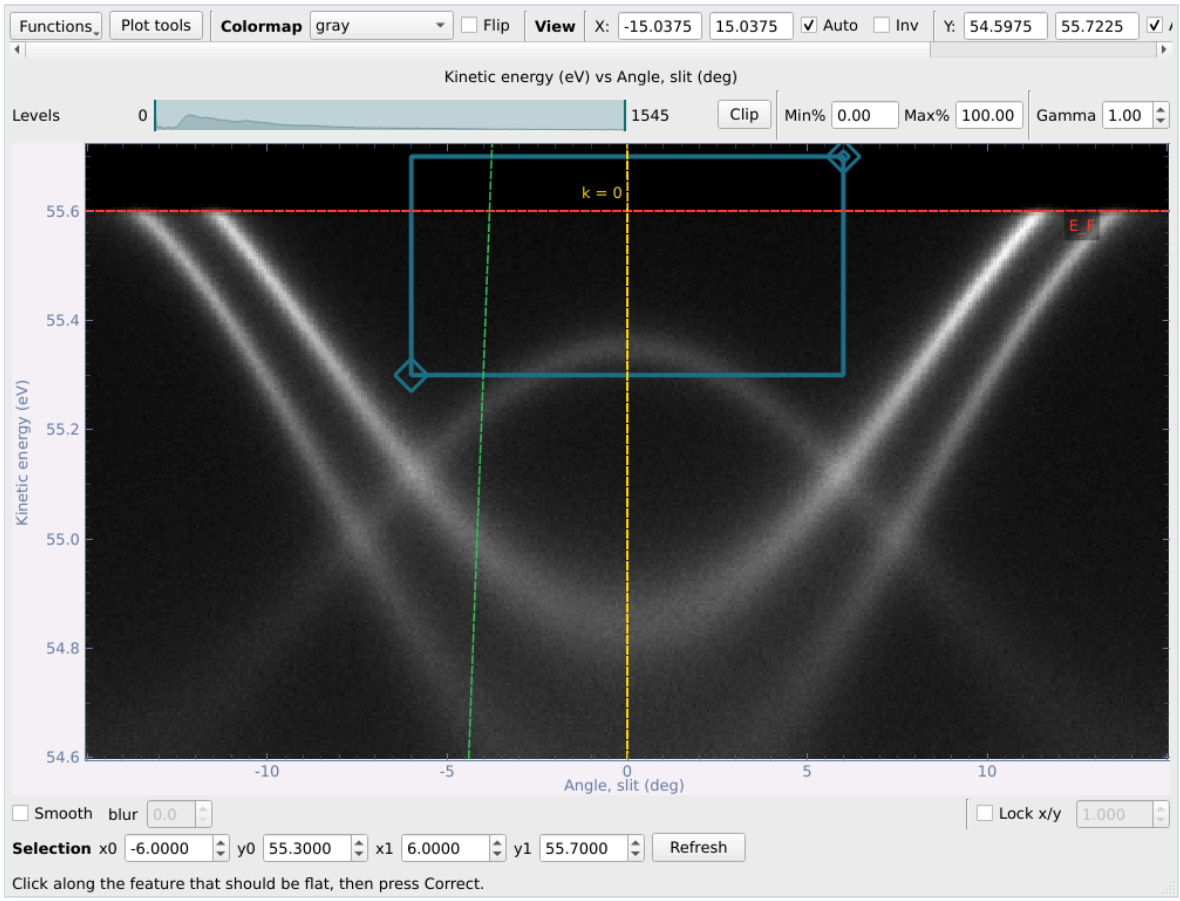
The **Functions** menu is organised in sections:

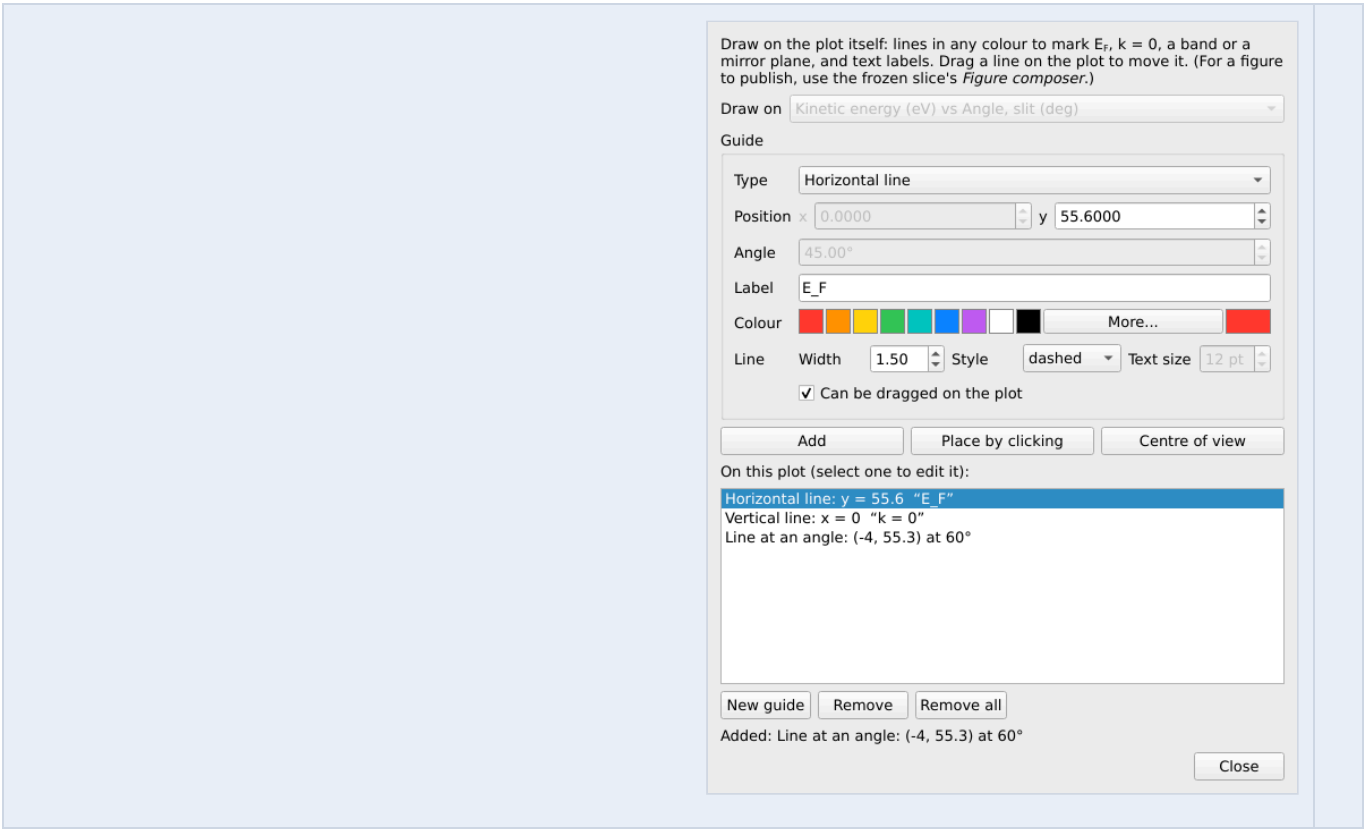
Viewer	Analysis	Transform	Visualization
map / kz map / k-map		Arbitrary cut, Convert to momentum (k), Align Fermi levels (kz map), Convert to momentum (kz, k∥), Remove detector grid	Brillouin zone (k-maps), Multi-slice figure
cut	Fermi edge fit, MDC / EDC fit	Straighten bands (FS correction), Convert to momentum (k), Cut arithmetic, Remove detector grid	
curve	Curve fit, Spin analysis	Crop, Bin, Normalise, Background, Shift axis, Add √N errors	Figure composer

An entry that does not apply is not shown; one that applies but cannot run yet is greyed, with the reason in its tooltip.

Plot tools: guide lines and labels on the plot

Plot tools on the top row of every viewer (cuts, maps, spatial scans, curves, frozen slices) opens a small window that draws **on the plot in front of you**:





- **Type**: *Vertical line*, *Horizontal line*, *Line at an angle* (through a point; the angle is in the plot's own units, i.e. the slope is tan(angle) y-units per x-unit), or *Text label*.
- **Position** (x, y), **Angle**, an optional **Label** (shown on the line, e.g. `E_F`, `Γ`), a **Colour** (nine one-click colours, or **More...**), **Width**, **Style** (solid, dashed, dotted, dash-dot) and **Text size**.

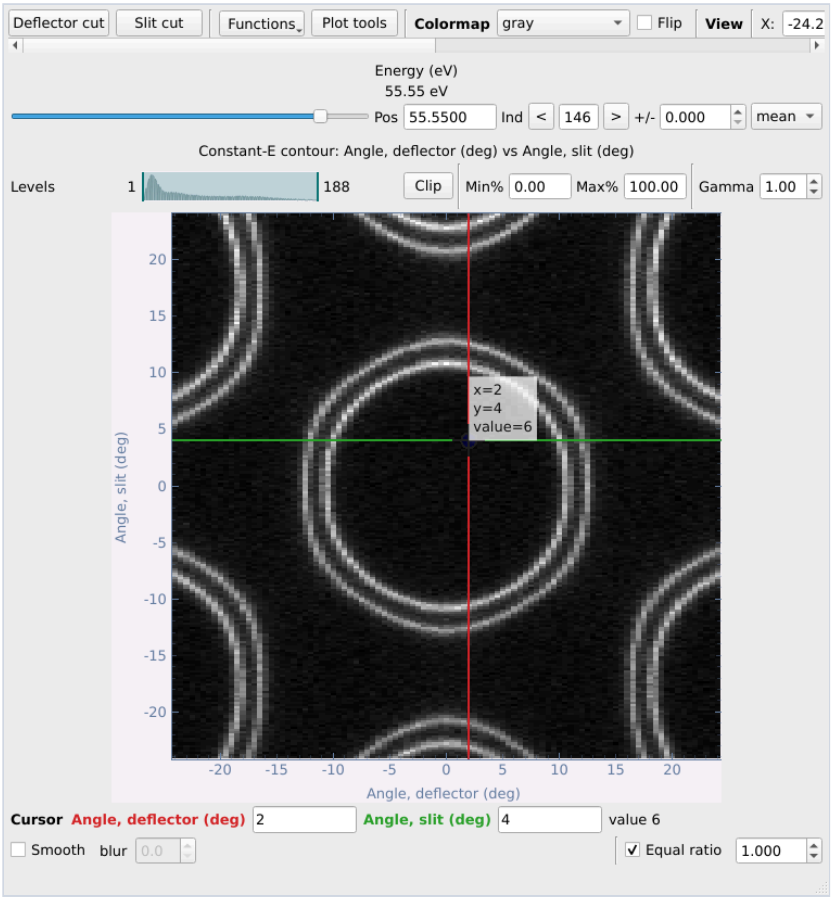
- **Add** draws it at the position typed (**Centre of view** fills that in); **Place by clicking** draws one wherever you click on the plot.
- **Drag** a line on the plot to move it; the window follows. Select a guide in the list to **edit** it — every change shows at once — or **Remove** it; **New guide** stops editing; **Remove all** clears the plot.
- With two panels (a spatial scan), **Draw on** chooses which.
- Guides stay while the slice moves (an E_F line stays at E_F as you scroll through a map) and while the Plot tools window is closed. The panel's **Export → Image** draws them too (*Include the Plot tools guides*).

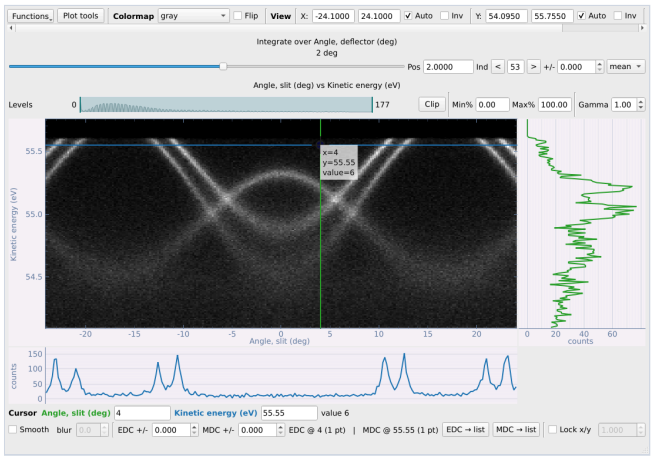
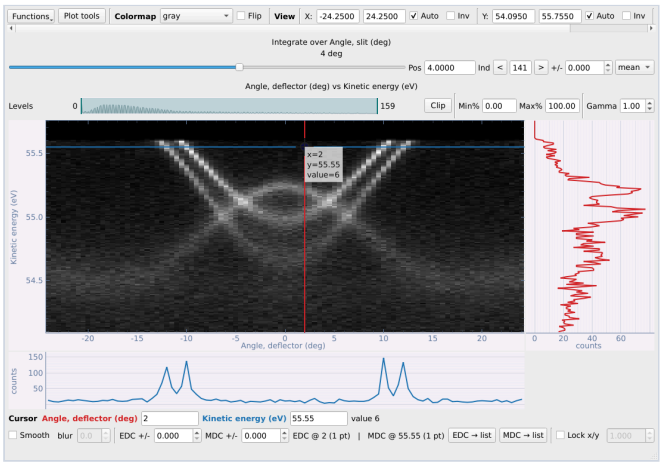
Plot tools are for looking at the data. To make a figure for a paper, use the **figure composer** and its **Figure tools** (§17).

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6. Maps: the contour and its two cuts

A map opens on its **constant-energy contour**. The slider at the top picks the energy (**Pos** in eV, or **Ind** as an index; **±** integrates over a window, **mean** or **sum**). **Deflector cut** and **Slit cut** open the two orthogonal cuts, each in its own window.



Slit cut (slit angle vs energy)	Deflector cut (deflector angle vs energy)
	

One cursor for all three windows. Switching the readout cursor on in any of them switches it on in the others, and moving it anywhere re-slices the others through the new point. The axes keep one colour each everywhere: **deflector red, slit green, energy blue**. A cursor line has the colour of the axis it holds constant, and the EDC or MDC it feeds is drawn in the same colour. Integration widths (the ± boxes) belong to each window and are never shared.

From the list you can also open a map's cut straight away (right-click → **Open slit cut** / **Open deflector cut**); the contour then stays hidden until you ask for it (*Functions → Show the map*).

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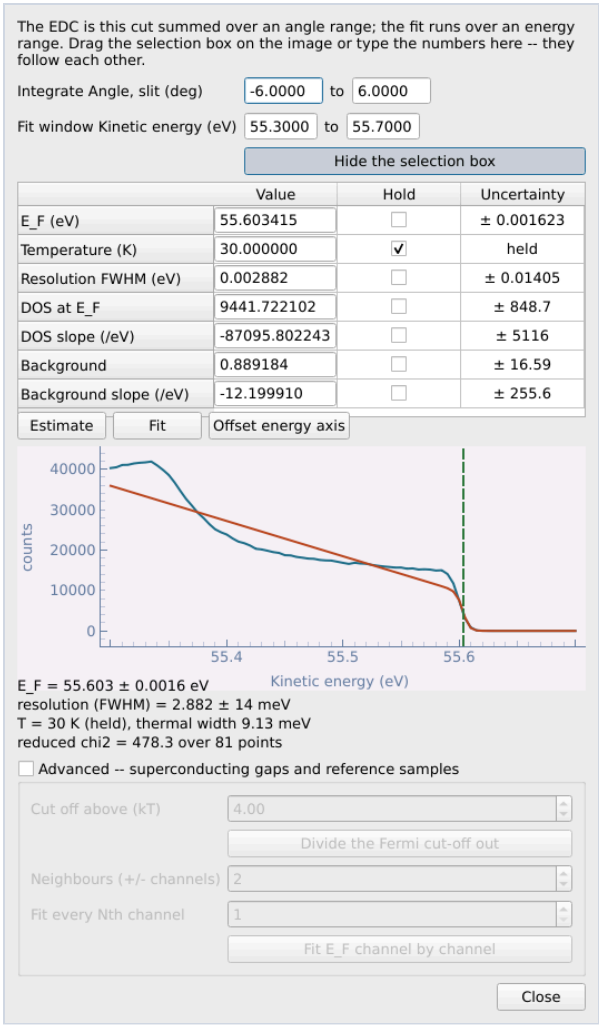
7. Cuts: from raw spectrum to band parameters

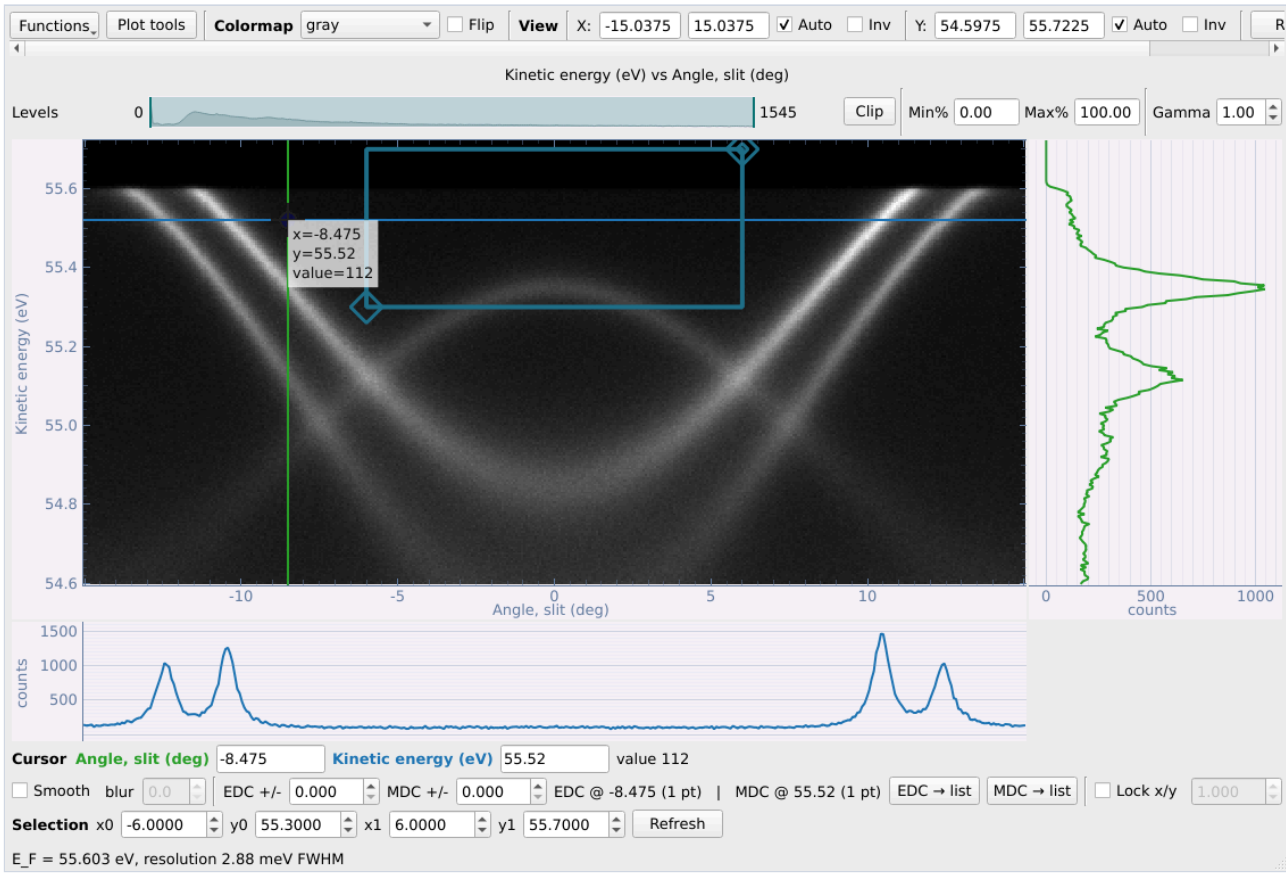
The usual order for a single cut is: **Fermi edge fit → offset the energy axis → convert to momentum → MDC/EDC fit**.

7.1 Fermi edge fit

Functions → Fermi edge fit... fits a Fermi edge (Fermi–Dirac × linear DOS, convolved with a Gaussian resolution) to the EDC summed over a box.

1. Press **Show the selection box** and drag the box on the cut over the edge (or type the angle and energy ranges). The angle range says which channels are summed; the energy range is the fit window.





2. **Estimate**, then **Fit**. Tick **Hold** to keep a parameter fixed (the temperature is held at the recorded value by default). Every parameter is reported with its uncertainty.

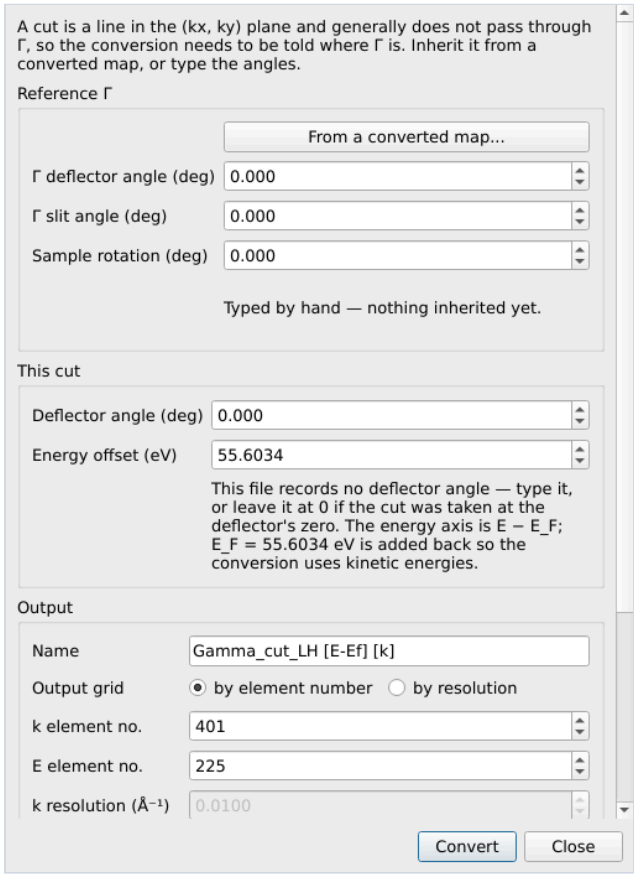
3. **Offset energy axis** lists a copy of the cut on **E − E_F**.

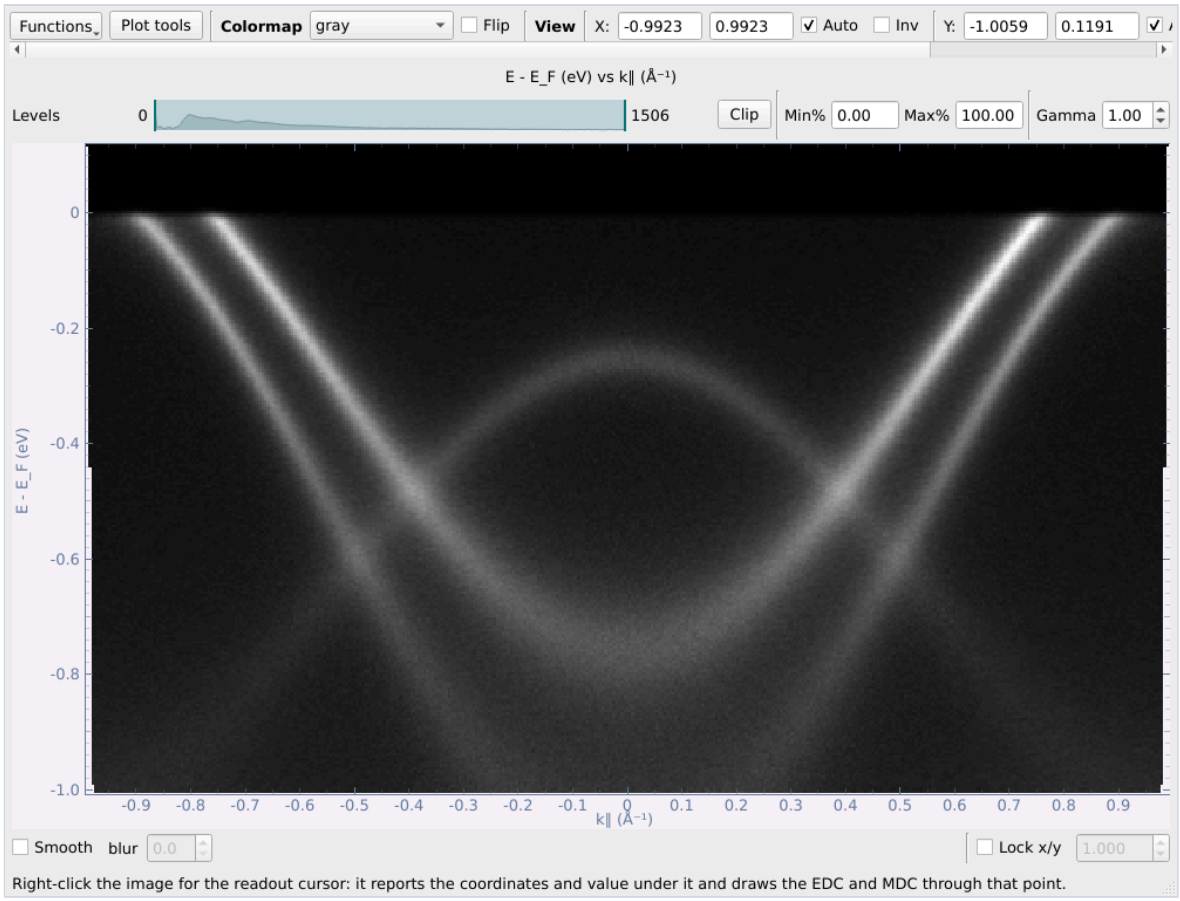
The *Advanced* corner divides the Fermi cut-off out (for gaps and reference samples) or fits E_F channel by channel.

7.2 Converting a cut to momentum

Functions → Convert to momentum (k)... converts the angle axis to k∥ (Å^{-1}). A cut is a line in the (kx, ky) plane that generally misses Γ, so the dialog asks where Γ is: type the angles, or **From a converted map...** to inherit them from a map already converted.

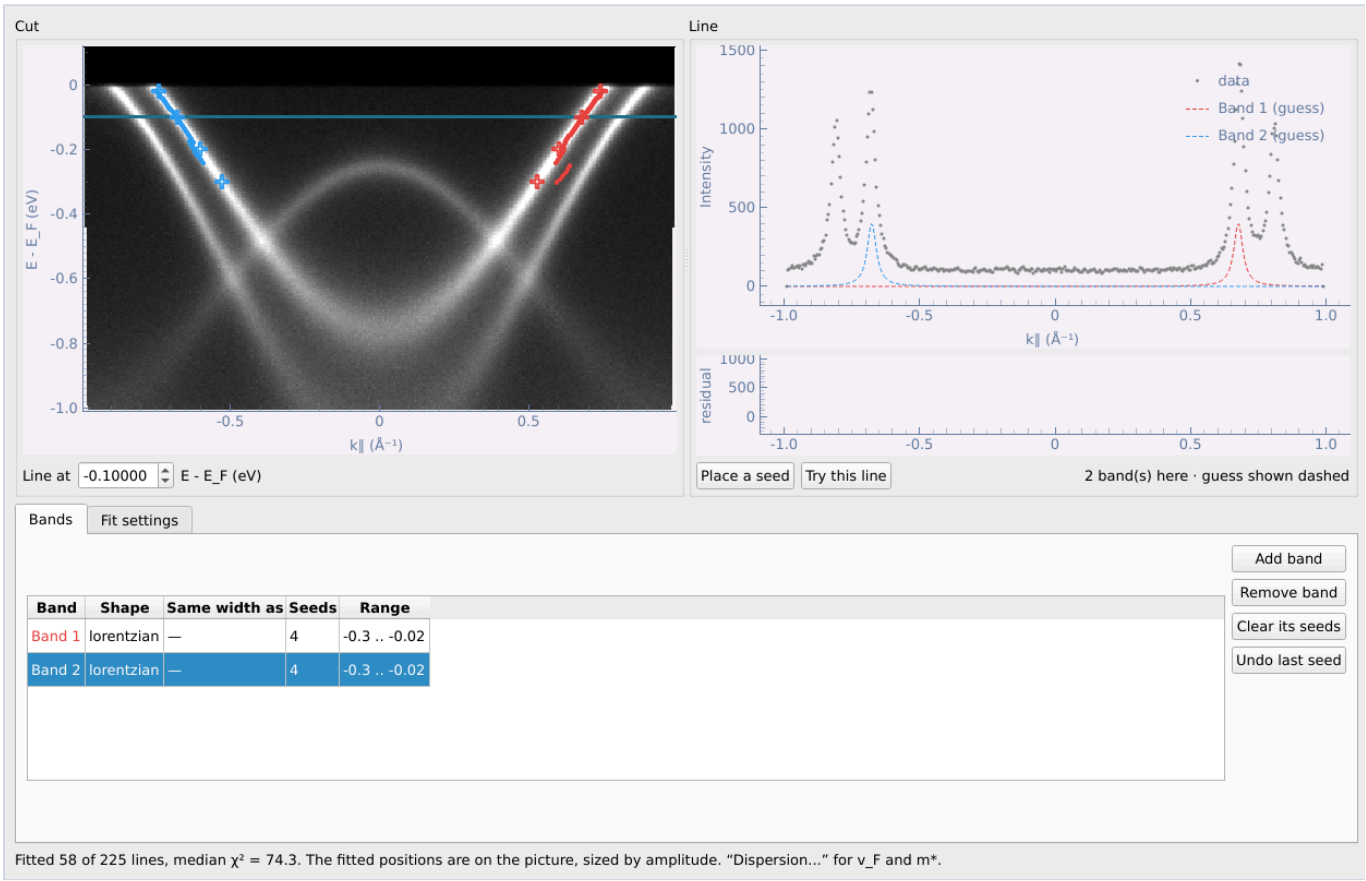
If the cut is already on E − E_F, the dialog fills in **Energy offset** with the E_F that was subtracted (the note under it says so): the conversion uses true kinetic energies, and the result stays on E − E_F.



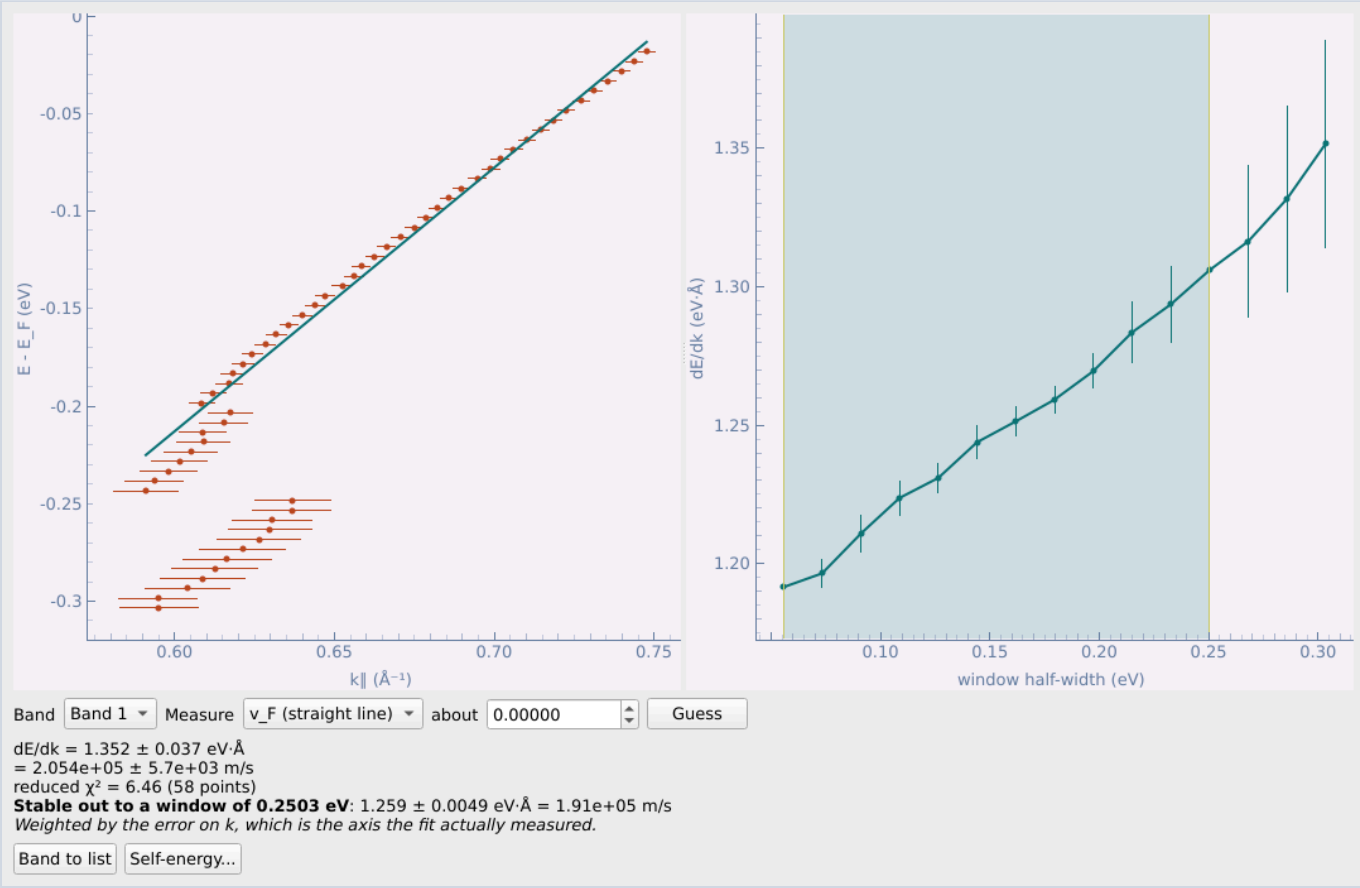


7.3 MDC / EDC fit

Functions → MDC / EDC fit... fits peaks line by line and reads the band off the fitted centres. It only accepts a cut whose axis is already in Å⁻¹ — a velocity in eV·deg would be meaningless.



1. **Bands** tab: **Add band** for each branch; select a band, press **Place a seed** and click the peak on the curve (right). Four or five seeds down a band are enough; **Line at** moves the line being looked at.
2. **Try this line** shows the fit of the current line (dashed: the guess).
3. **Fit settings** tab: MDC or EDC, peak shape (Lorentzian, Gaussian, Voigt), background, step, then **Fit the series**. Fitted positions are drawn on the image, sized by amplitude.
4. **Dispersion...** fits the band: **v_F** (straight line), **m*** (parabola) — and checks how far out the fit is stable. **Self-energy...** gives Re Σ and Im Σ.

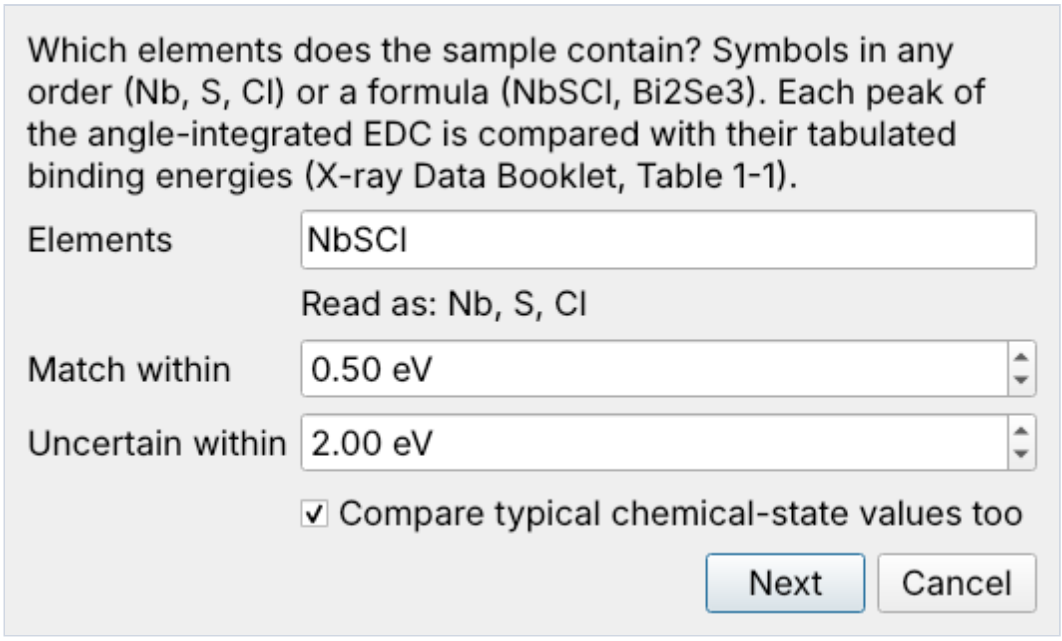


Save table... writes every fitted line; **Band to list** puts a band in the main list.

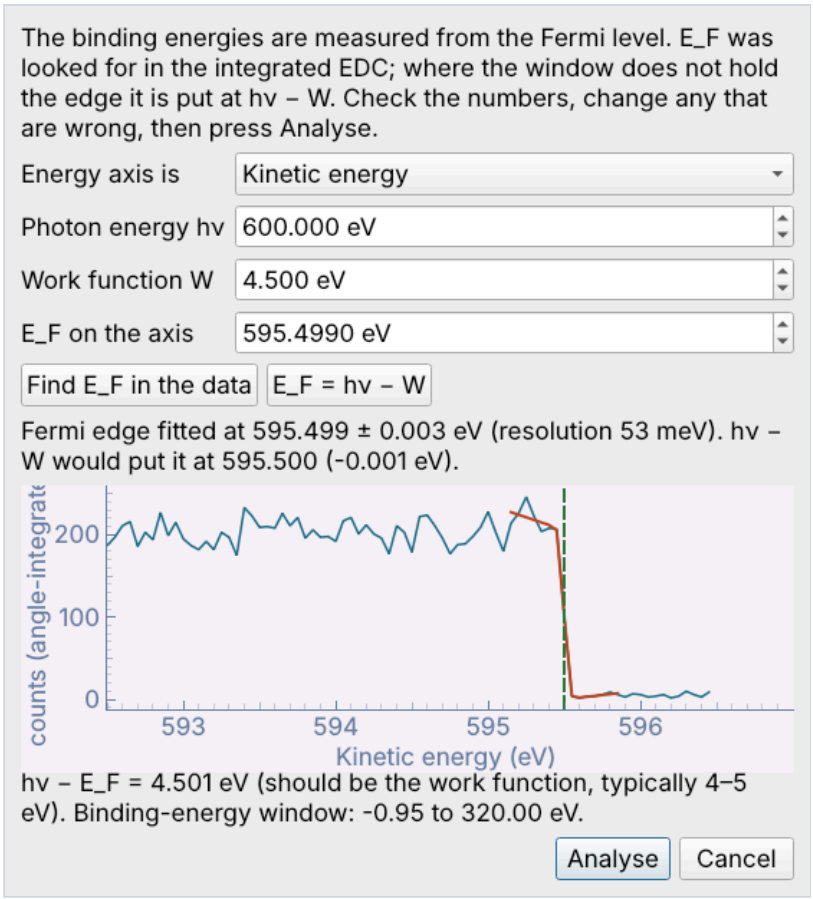
7.4 Core-level check

A wide scan of the core levels is listed as a cut too. Right-click it in the main list → **Core-level check...** to find out what its peaks are.

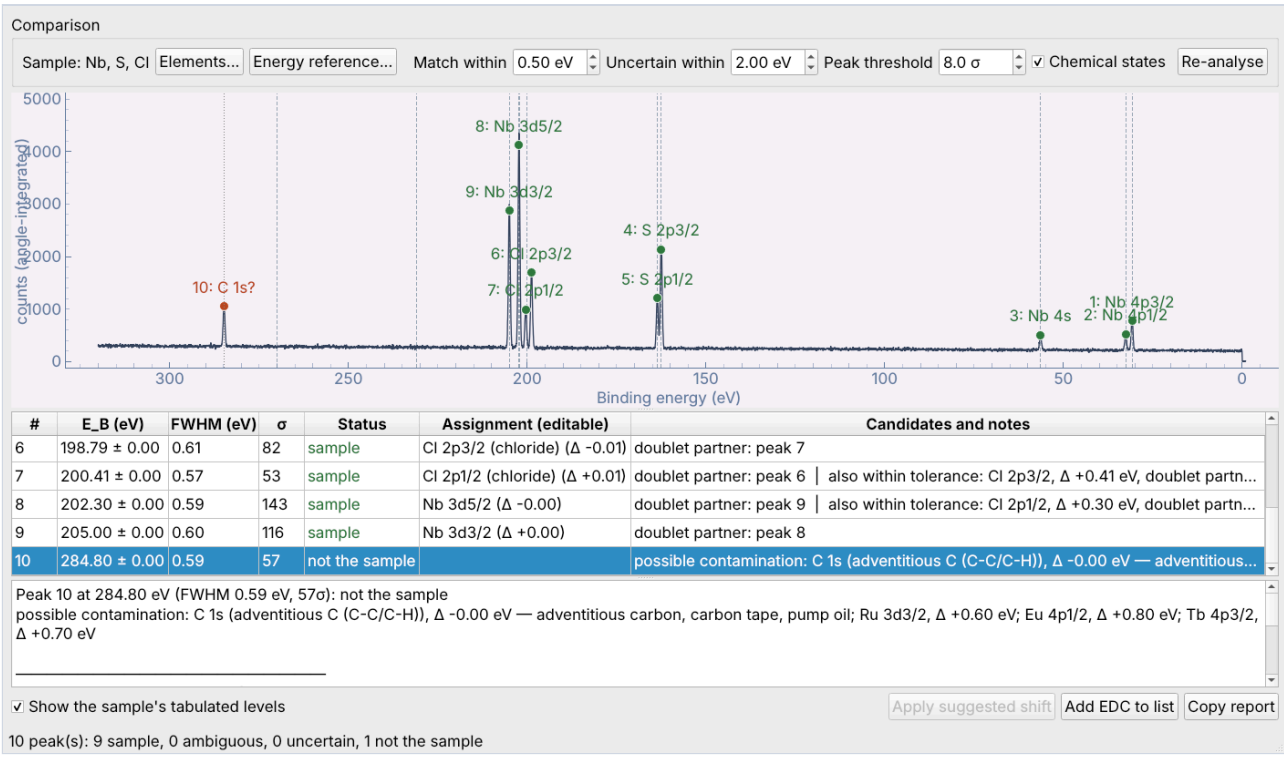
1. Type the sample's elements — `NbSCl` or `Nb, S, Cl` — and press **Next**. *Match within* (0.5 eV) is how close a peak must be to a tabulated level to count as the sample's.



2. Check the energy reference. E_F is fitted when the window holds the Fermi edge, otherwise put at hν − W. Change the photon energy, the work function or E_F if they are wrong (the green line can be dragged); a hν − E_F far from 4–5 eV means the photon energy is off. Press **Analyse**.



3. The results: the EDC on a binding-energy axis, each peak marked — green *sample* (orbital named), orange *uncertain* (a level of the sample within 2 eV: chemical shift or charging — your call), red *not the sample* (common elements that could explain it: contamination, higher-order light, Auger lines). Click a row for the full list of candidates. Type your own assignment in the *Assignment* column.



Weak lines and peaks by hand. Tick **Log counts** to see weak lines next to a strong one; **Min height** (default 0.2 % of the strongest) sets how small a found peak may be. To add a peak the search missed, press **Click to add peak** and click the plot (a double-click always works), or type its binding energy after *or at* and press **Add**: its position is refined by a local fit, it is marked ◆, and it is named with every tabulated level within ±2 eV. **Remove selected** drops peaks; **Reset peaks** brings back the found ones and forgets those added by hand. Assignments you typed survive a **Re-analyse**.

Peak table... lists every tabulated level, filtered by element and binding-energy range: look a peak up, show a level on the plot, or add it as a peak with that level as its assignment. **Select range** shows a band you drag over a region; **Zoom window...** opens it (or the current view) in a window of its own with a cursor, a Gaussian fit at the cursor, and the negative second derivative divided by the intensity — which shows a shoulder or a weak line beside a strong one as clearly as the strong one.

If every peak is off by the same amount, the notes say so and **Apply suggested shift** corrects it (a charged sample, or a wrong E_F). **Add EDC to list** lists the EDC on the binding-energy scale; **Copy report** copies the whole result as text. Measuring at a second photon energy settles what stays doubtful: core levels stay put in binding energy, Auger lines and higher-order light move.

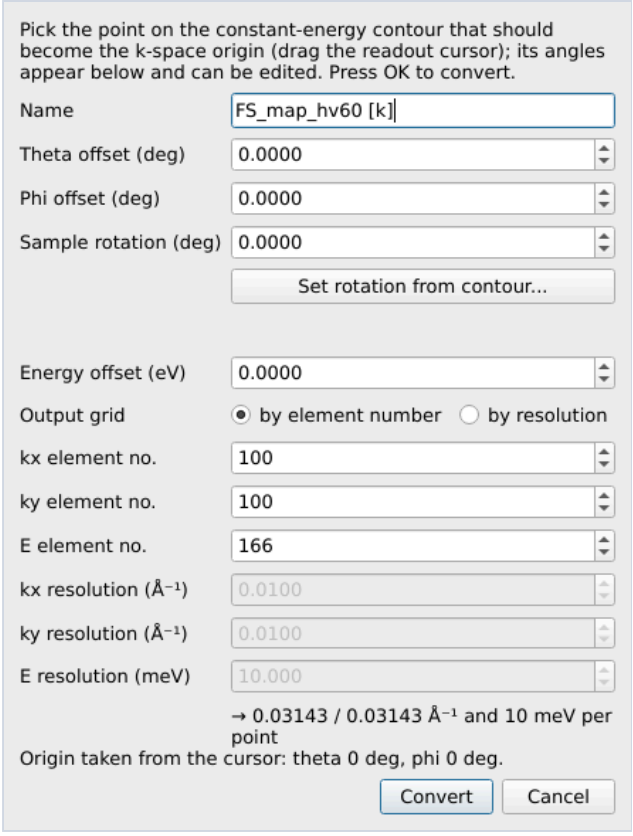
---

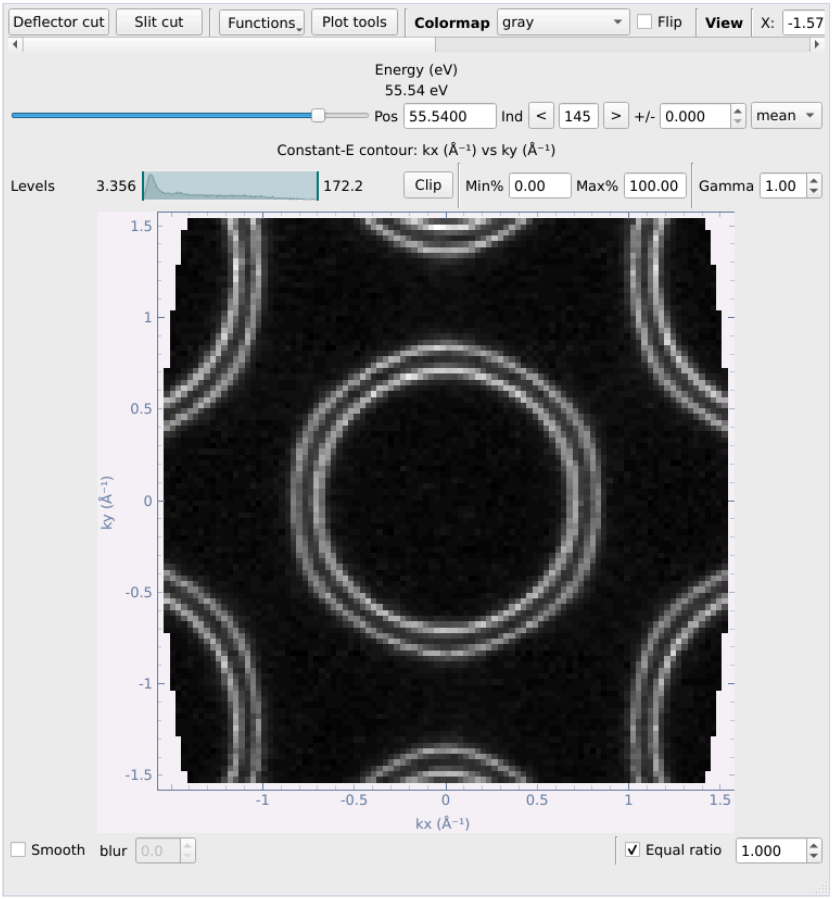
8. From angles to momentum for a map, and the Brillouin zone

8.1 Converting a map to momentum

Functions → Convert to momentum (k)... on the contour window. Drag the readout cursor to the point that should become **k = (0, 0)** — the dialog follows it — then **Convert**. The result is a new `k-map` row in the list.

- **Sample rotation** turns the map about the surface normal; **Set rotation from contour...** reads it off a direction you click on the contour (two points along a mirror line, say).
- **Output grid**: number of points, or resolution (Å⁻¹, meV).
- **Energy offset** is added to the energy axis before converting, for an axis that is not a true kinetic energy (filled in automatically for a map on E − E_F).

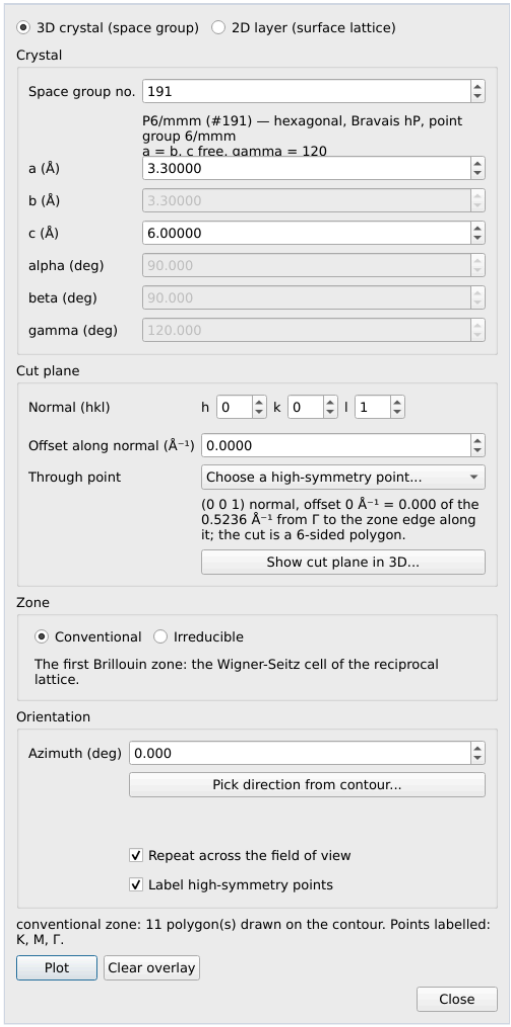


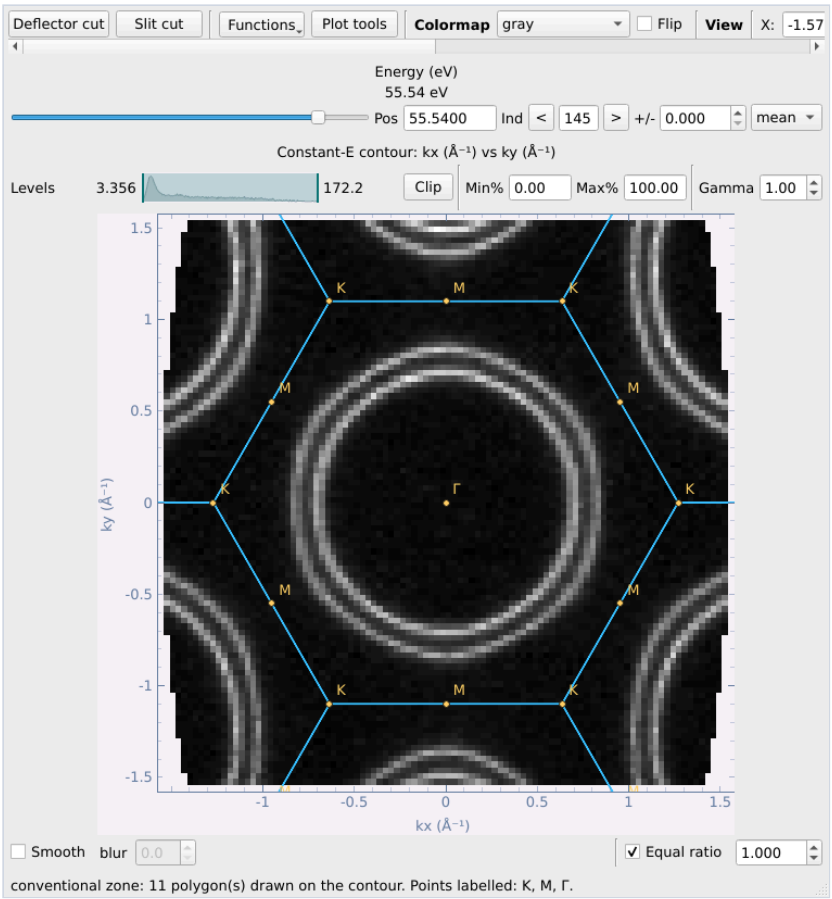


8.2 Brillouin zone

On a k-map: *Functions → Visualization → Brillouin zone...*.

1. **Space group** first (1–230); the lattice parameters it fixes are greyed and follow the free ones. Or switch to **2D layer** for a surface lattice.
2. **Cut plane**: the surface normal (hkl) and an offset along it — the cut of the 3-D zone by that plane is drawn, for any surface, not only (001). **Through point** lists the offsets at which the plane passes through a high-symmetry point (Γ, Z, L, F … for a rhombohedral crystal); choose one rather than working the number out. **Show cut plane in 3D…** previews it, with the named points marked and the ones in the plane ringed; **Label high-symmetry points** writes them on the contour, which is the quickest way to set the azimuth. The dialog opens on the last crystal you used, so a second map of the same sample needs no retyping.
3. **Conventional** or **Irreducible** zone; **Azimuth** (or **Pick direction from contour…**); **Repeat across the field of view**.
4. **Moiré** adds a second, twisted or mismatched layer and its moiré zone.
5. **Plot**.

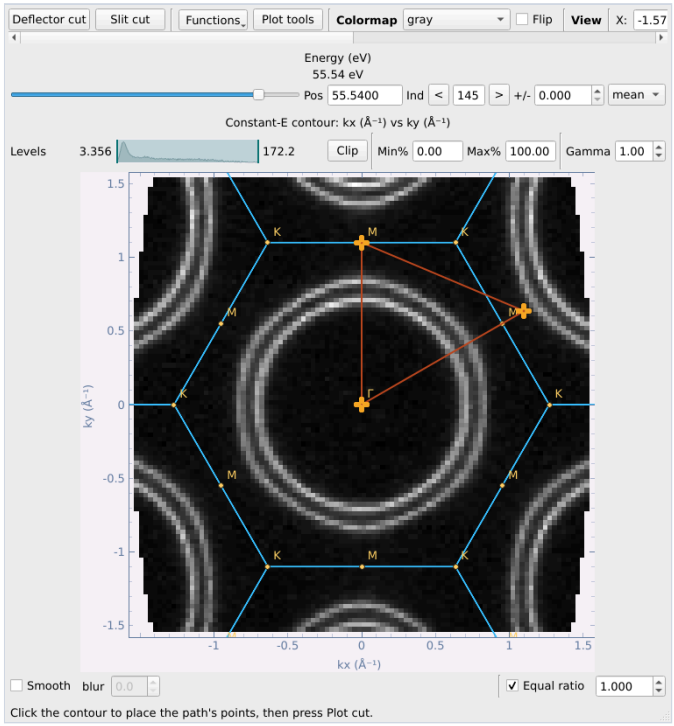
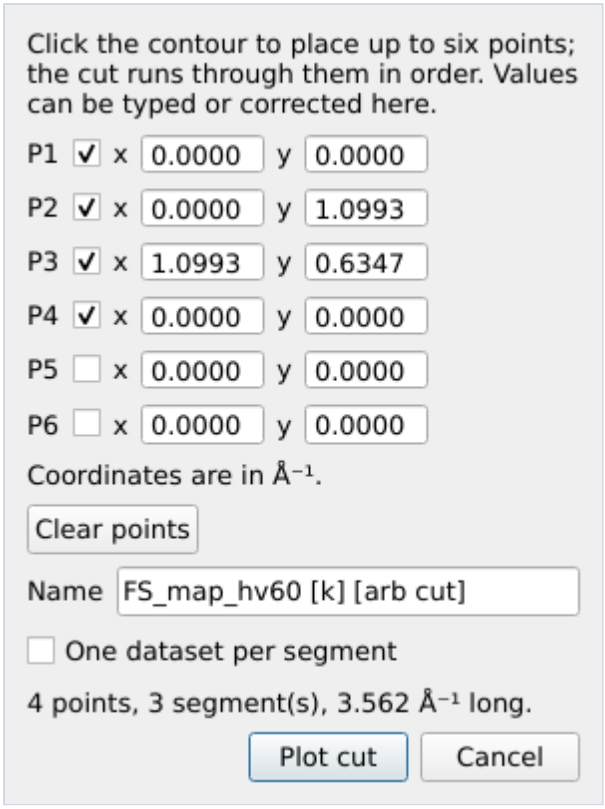




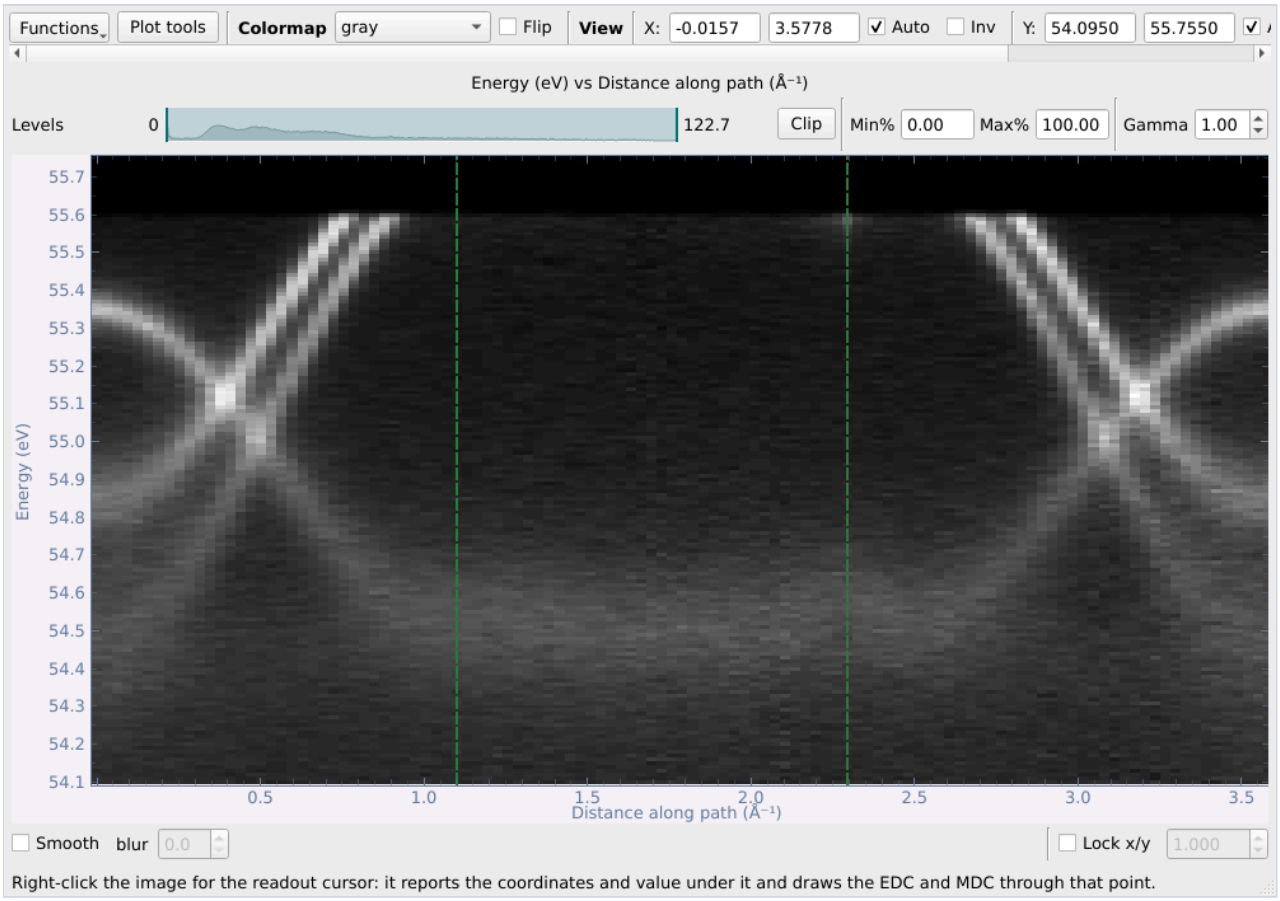
9. Arbitrary cuts and Fermi-surface correction

9.1 Arbitrary cut

Functions → Arbitrary cut… on any contour: click up to six points (or type them) and the cut runs through them in order — here Γ → M → K → Γ on the k-map. **One dataset per segment** lists each leg separately.

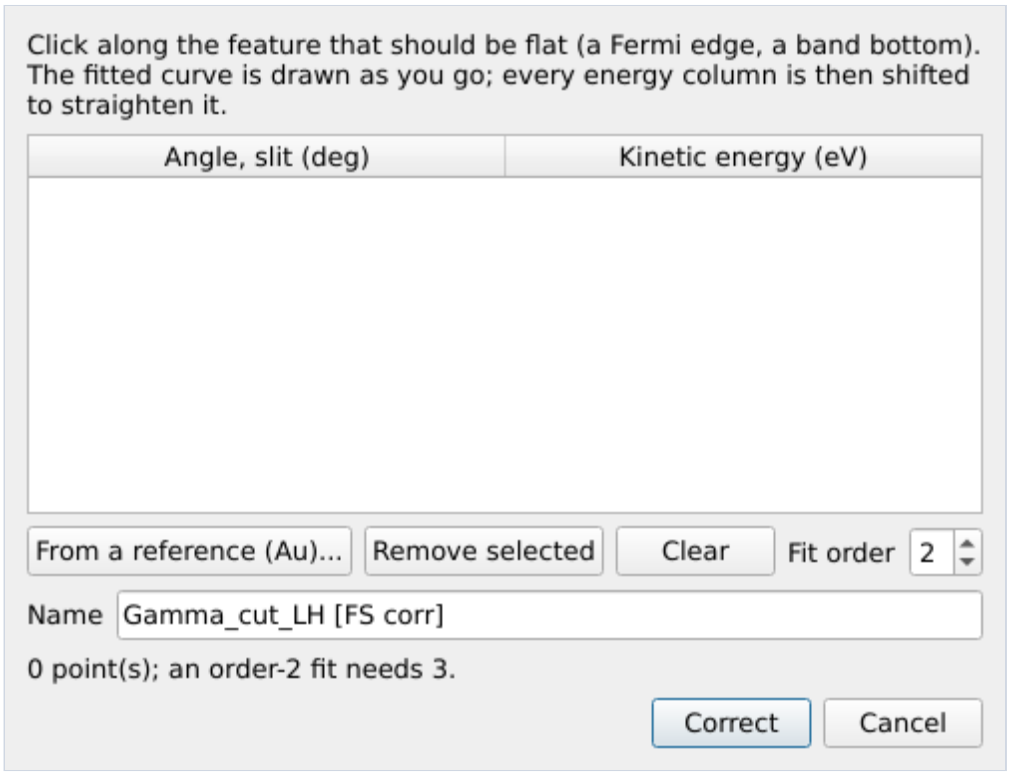
Path on the contour	Points
	

The result is a cut against the distance along the path; the corners of the path are marked with dashed green lines, so a kink there is not mistaken for physics.



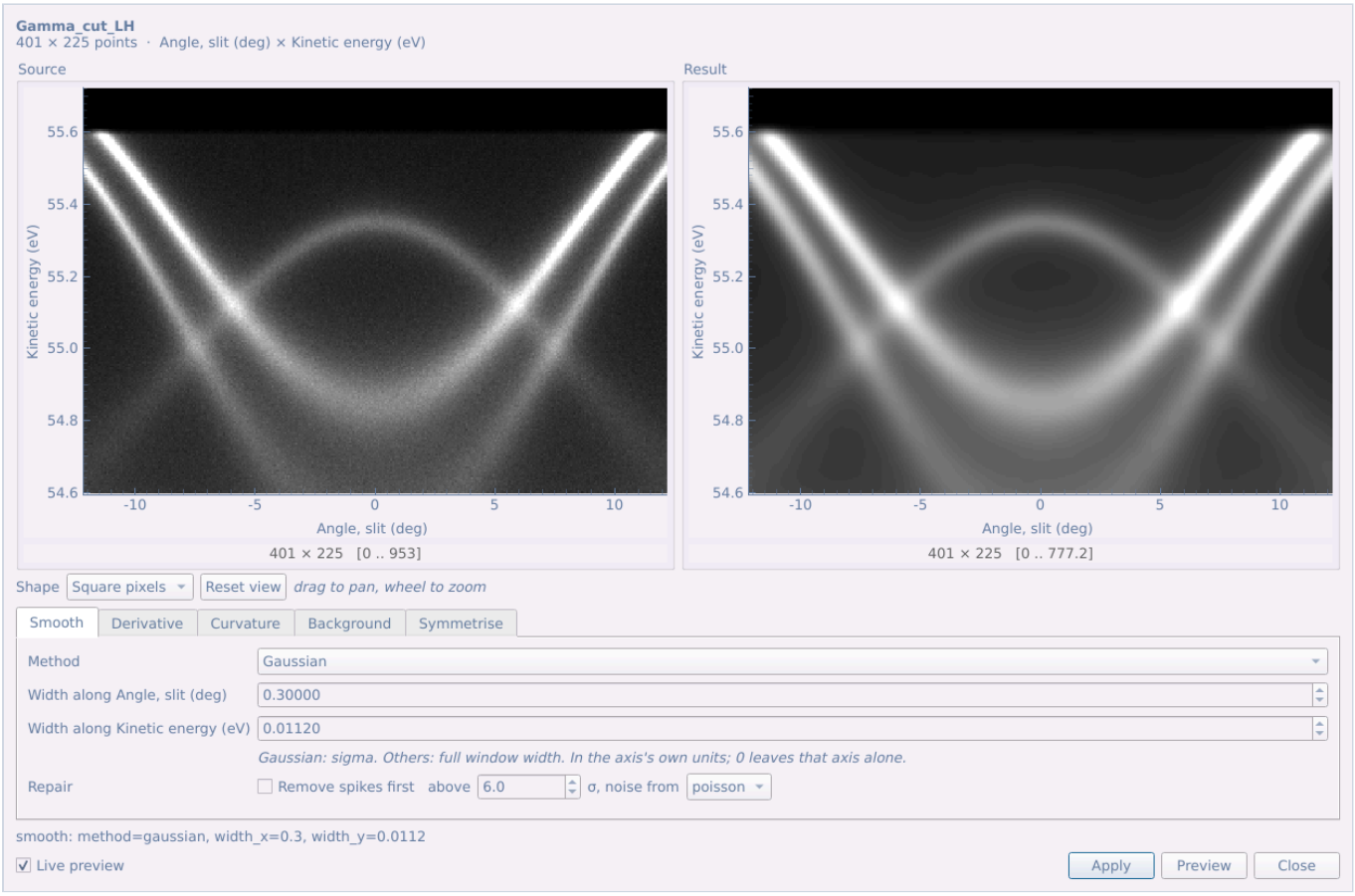
9.2 Fermi-surface (FS) correction

Functions → Straighten bands (FS correction)... on a cut (or a map's slit cut) straightens a feature that should be flat — usually the Fermi edge, which curves across the analyser angle. Click along the feature; a polynomial (order 2 by default) is fitted and every angle column is shifted in energy. **From a reference (Au)...** takes the curve from a gold spectrum in the list instead. On a map's slit cut the same shift is applied to the whole cube.



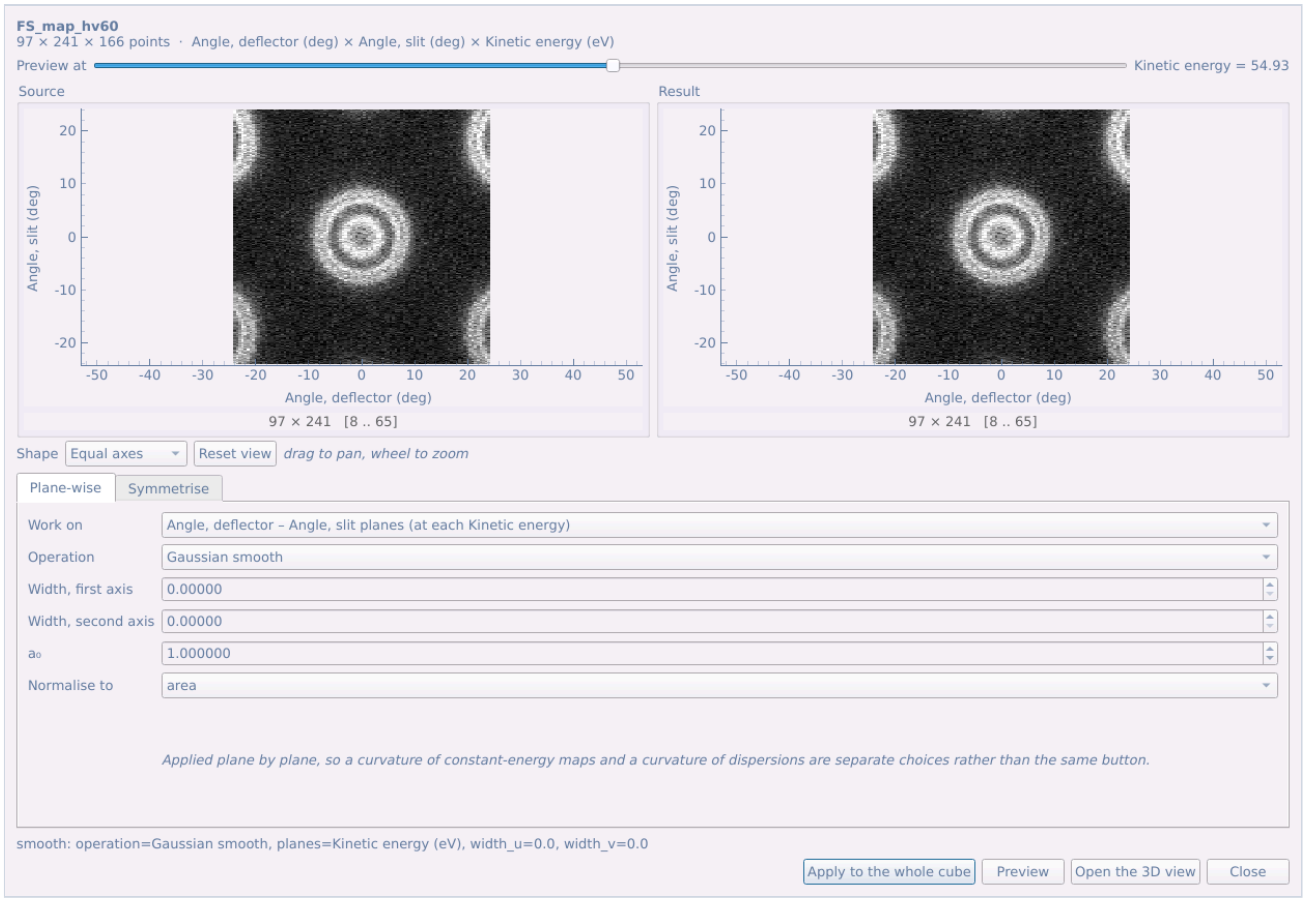
10. Processing: smoothing, derivatives, curvature

Select one or more cuts (or maps) in the list and press **Smooth / derivative...**. Every result is a new dataset carrying the steps that made it.

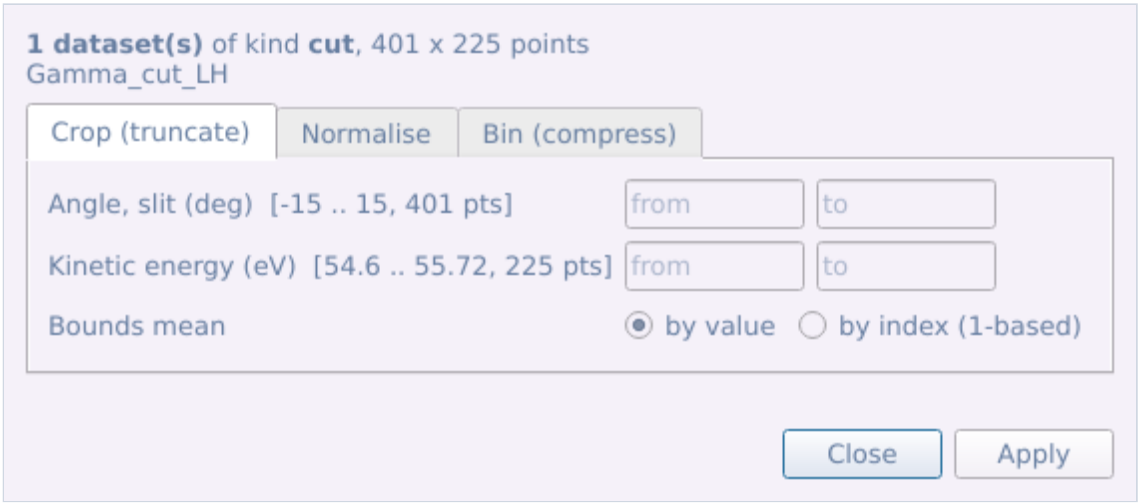


- Tabs: **Smooth** (Gaussian, box, …; optional despiking), **Derivative**, **Curvature** (1-D and 2-D), **Background**, **Symmetrise**.
- **Source** and **Result** are shown side by side on the same scale, so you can see when a setting has eaten the band. They pan and zoom together.
- **Shape**: *Square pixels* (one data point is square — undistorted for an angle-vs-energy cut), *Equal axes* (one unit is the same length on both axes — right for angle-vs-angle or k-vs-k; chosen automatically when both axes have the same unit), or *Fill the box*.
- **Live preview** re-computes as you change a value; **Apply** lists the result.

For a map the panel works plane by plane, and asks **which planes** first — constant-energy maps or dispersions — since the curvature of a contour and the curvature of a cut are different quantities. **Preview at** picks the plane shown; **Apply to the whole cube** runs it on every plane.

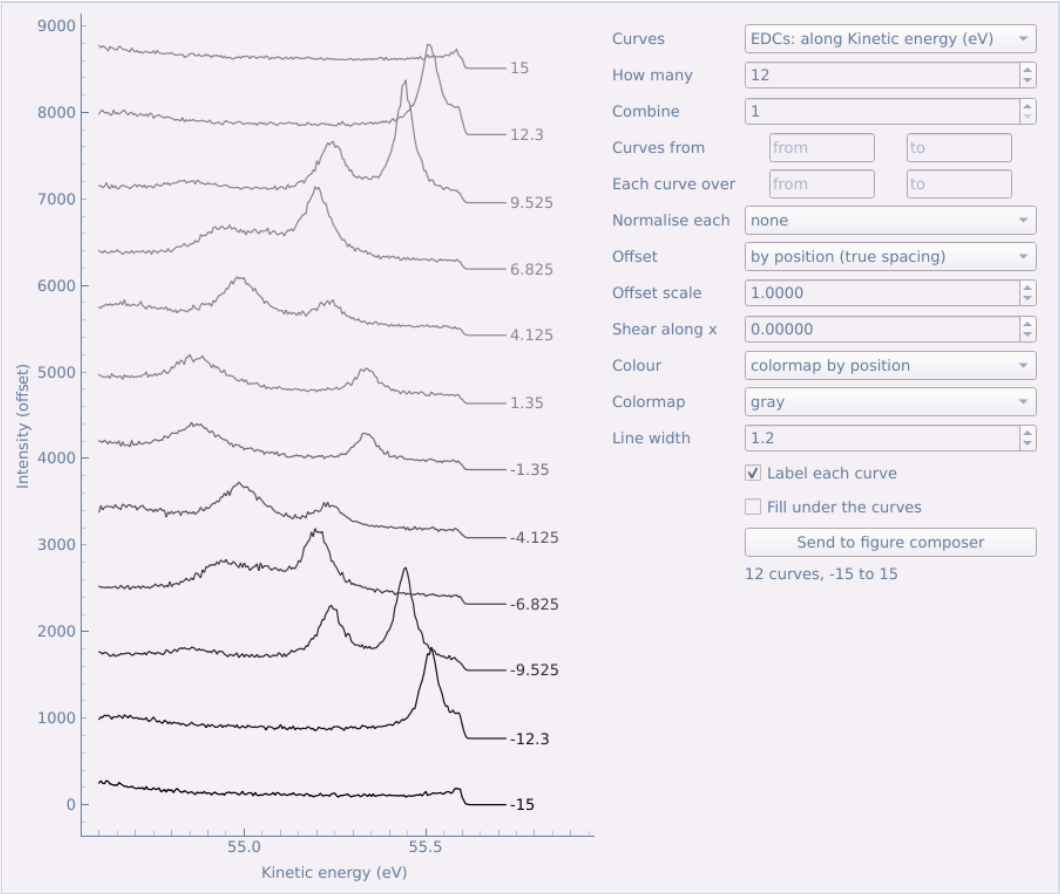


Crop / bin / normalise... (main window) changes the grid or the scale of one or several datasets at once — as long as they have the same kind and the same number of points on every axis. Tabs: **Crop (truncate)** to a range, **Normalise** each spectrum to itself, **Bin (compress)** points together.

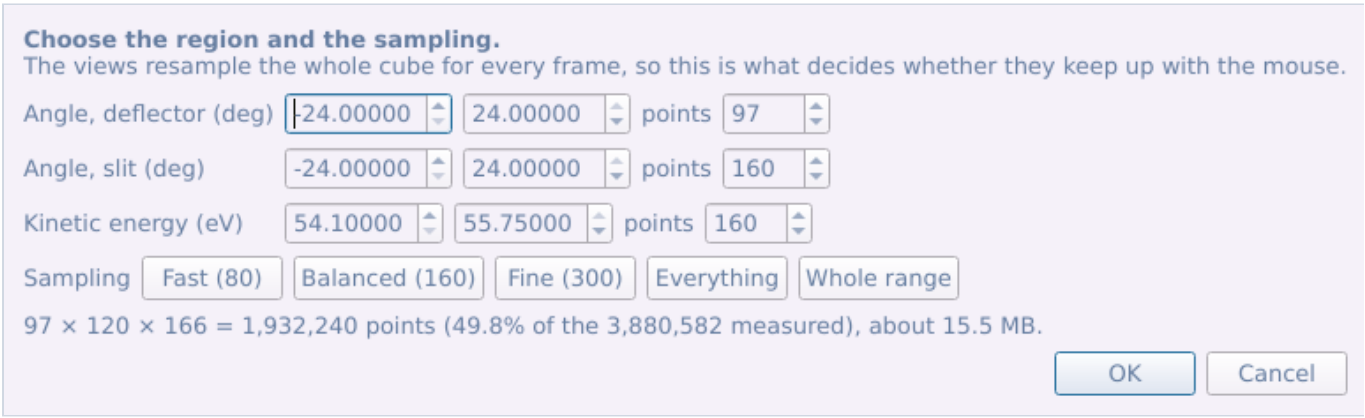


11. Stack plots and the 3-D view

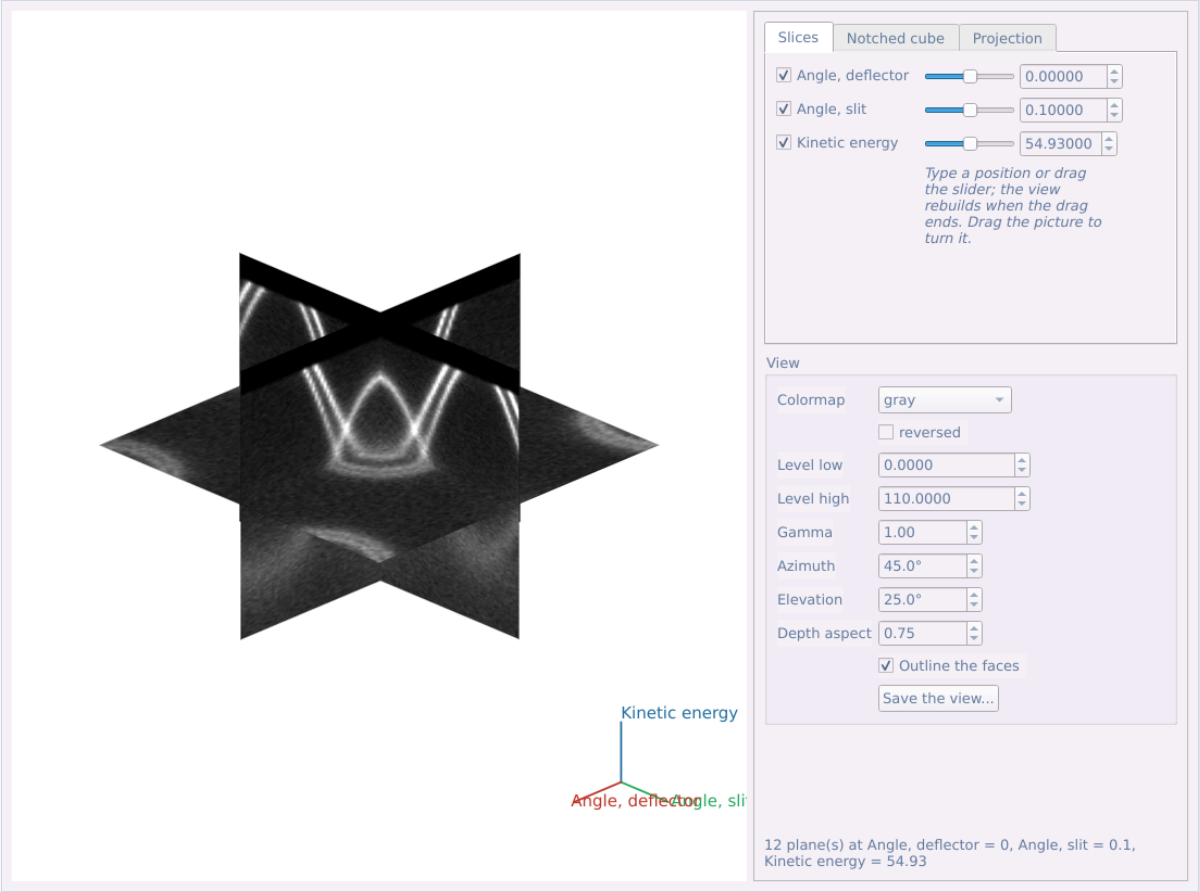
Stack plot (right-click a cut): a waterfall of EDCs or MDCs, with the number of curves, binning, normalisation, offset, shear and colouring controls; **Send to figure composer** makes it a figure panel.



3D view (right-click a map) first asks for the region and the sampling, since every frame resamples the cube — *Fast*, *Balanced* or *Fine* decide whether it keeps up with the mouse.

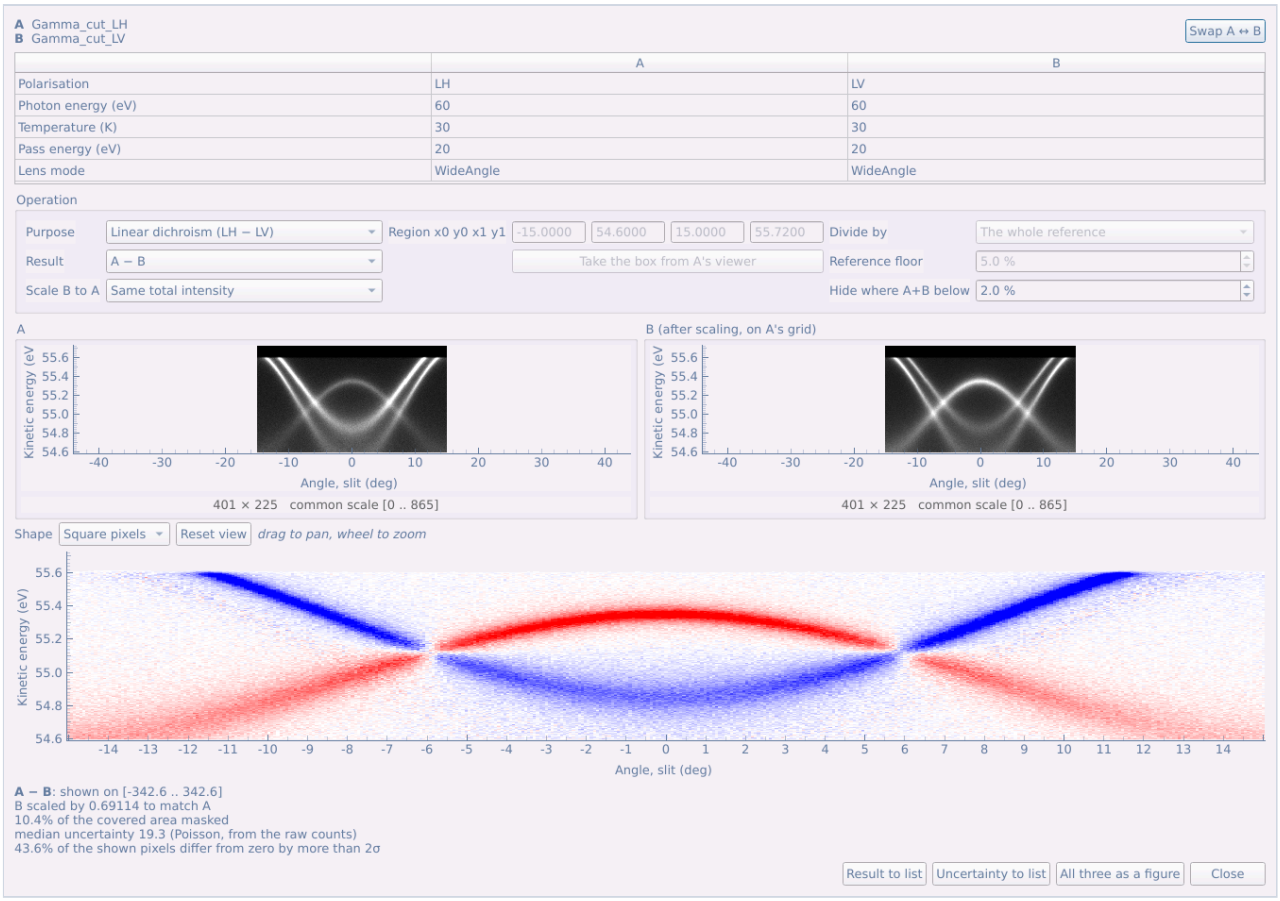


Then drag the picture to turn it. **Slices** shows three orthogonal planes at the positions you set, **Notched cube** cuts a corner out of the cube, and **Projection** sums along an axis. Colormap, levels, gamma, azimuth, elevation and depth aspect are on the right; **Save the view...** exports the picture.



12. Cut arithmetic: dichroism and reference division

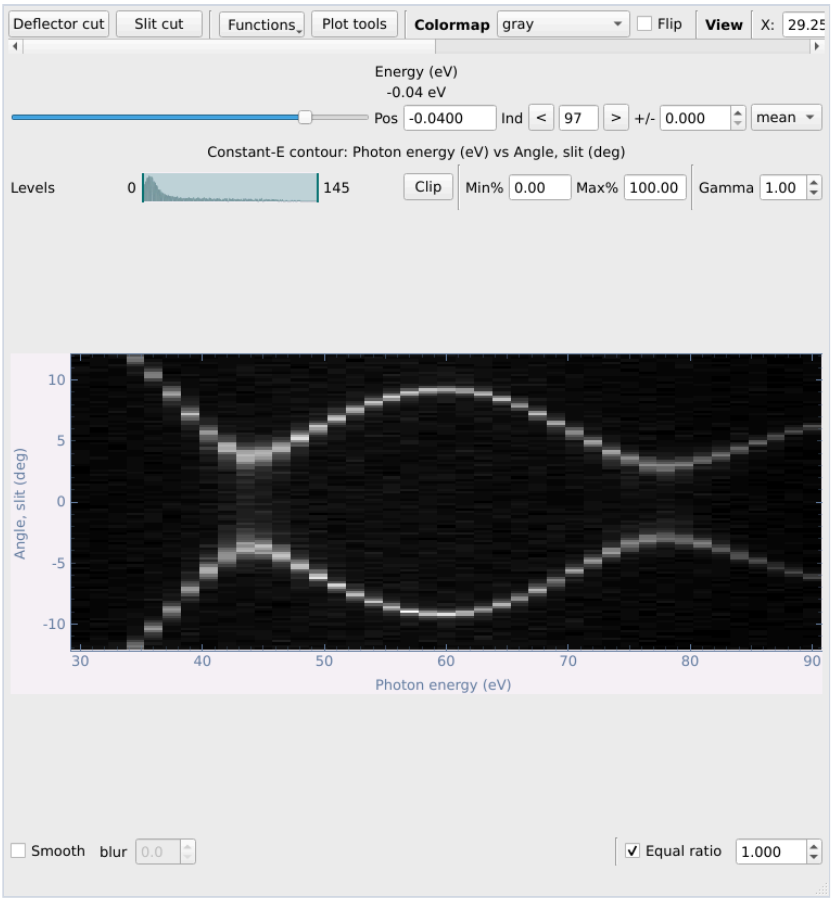
Two cuts combined into a third: linear dichroism (LH − LV), circular dichroism, dividing by a gold reference, or plain differences, ratios and sums. Select two cuts in the list → right-click → **Cut arithmetic on the two...** (the first selected is A), or in a cut's viewer *Functions* → *Cut arithmetic...* and then click the other cut in the list.



- **The table at the top** compares what the two measurements recorded (polarisation, photon energy, angles, sample position, temperature, pass energy, lens mode). Rows in orange differ beyond a tolerance — a dichroism map between two spots on the sample is a map of *that* difference.
- **Purpose** presets: *Linear dichroism* (A − B, B scaled to the same total intensity), *Circular dichroism* ((A − B)/(A + B)), *Divide by a reference*, *Custom*. Warnings appear if the polarisations look swapped (**Swap A ↔ B**).
- **Scale B to A** by total intensity or by a region; **Hide where A+B below** blanks the background.
- A and B are drawn on one colour scale; the result on a red-white-blue scale symmetric about zero. With raw counts on both, the Poisson uncertainty is propagated and summarised under the plot.
- **Result to list**, **Uncertainty to list**, **All three as a figure**.

13. Photon-energy scans (kz)

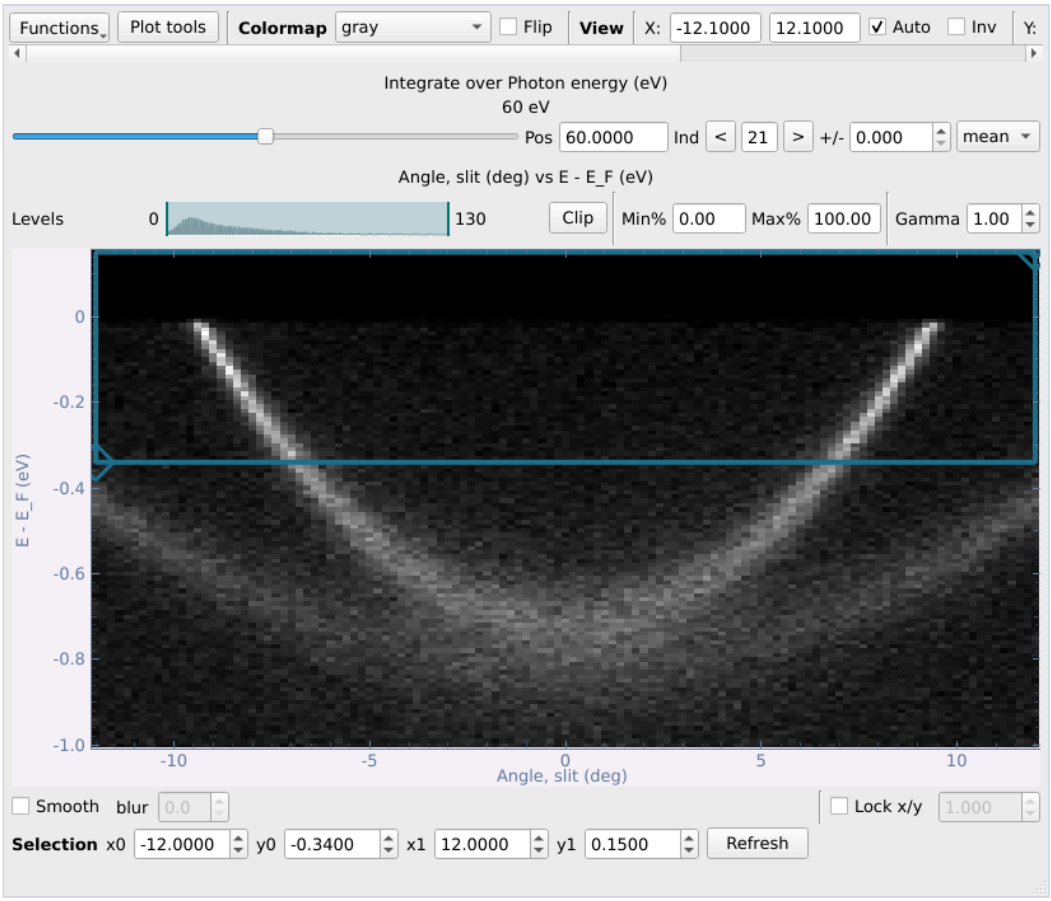
A photon-energy scan is listed as a `kz map`; its contour is photon energy against angle at one energy.



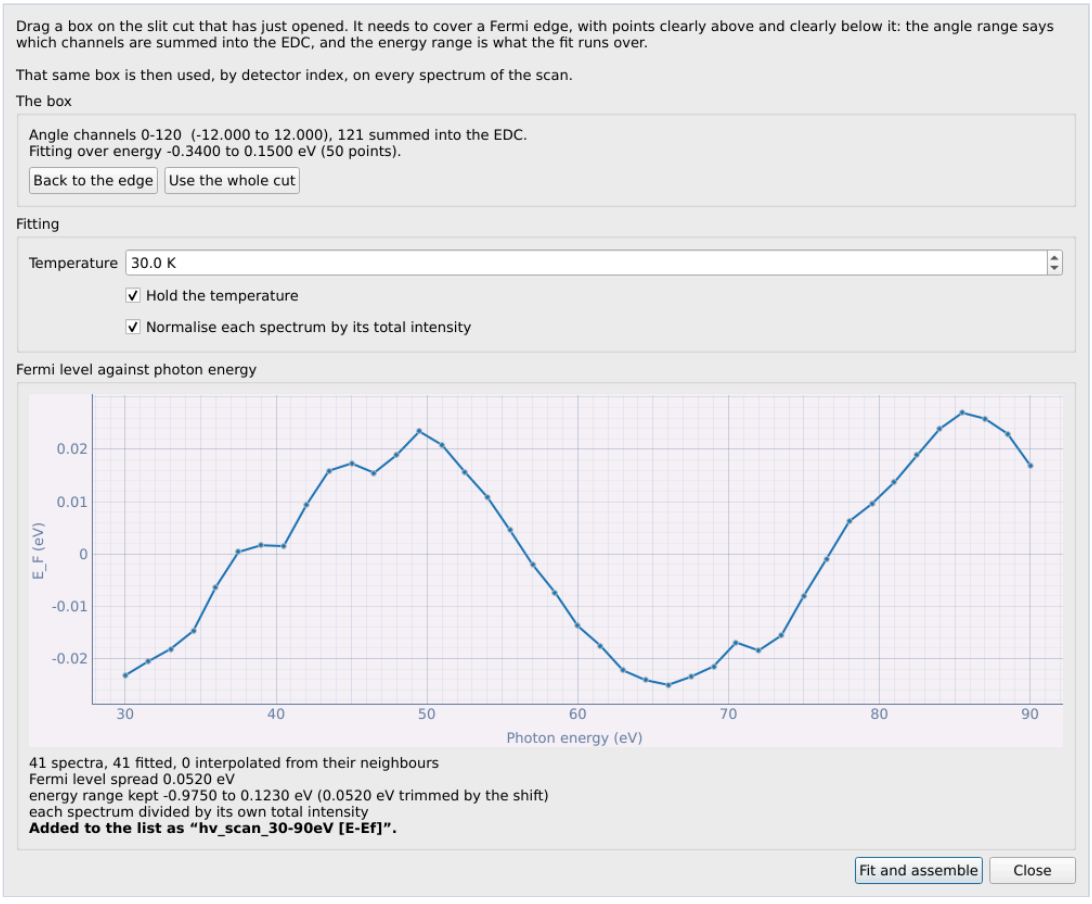
13.1 Aligning the Fermi levels of a kz map

Nothing in the files says where each spectrum's Fermi level really is. *Functions → Align Fermi levels (kz map)...* measures it from the spectra themselves.

1. The slit cut opens with a box on it. Drag the box over a Fermi edge — the angle range sets the channels summed, the energy range the fit window.



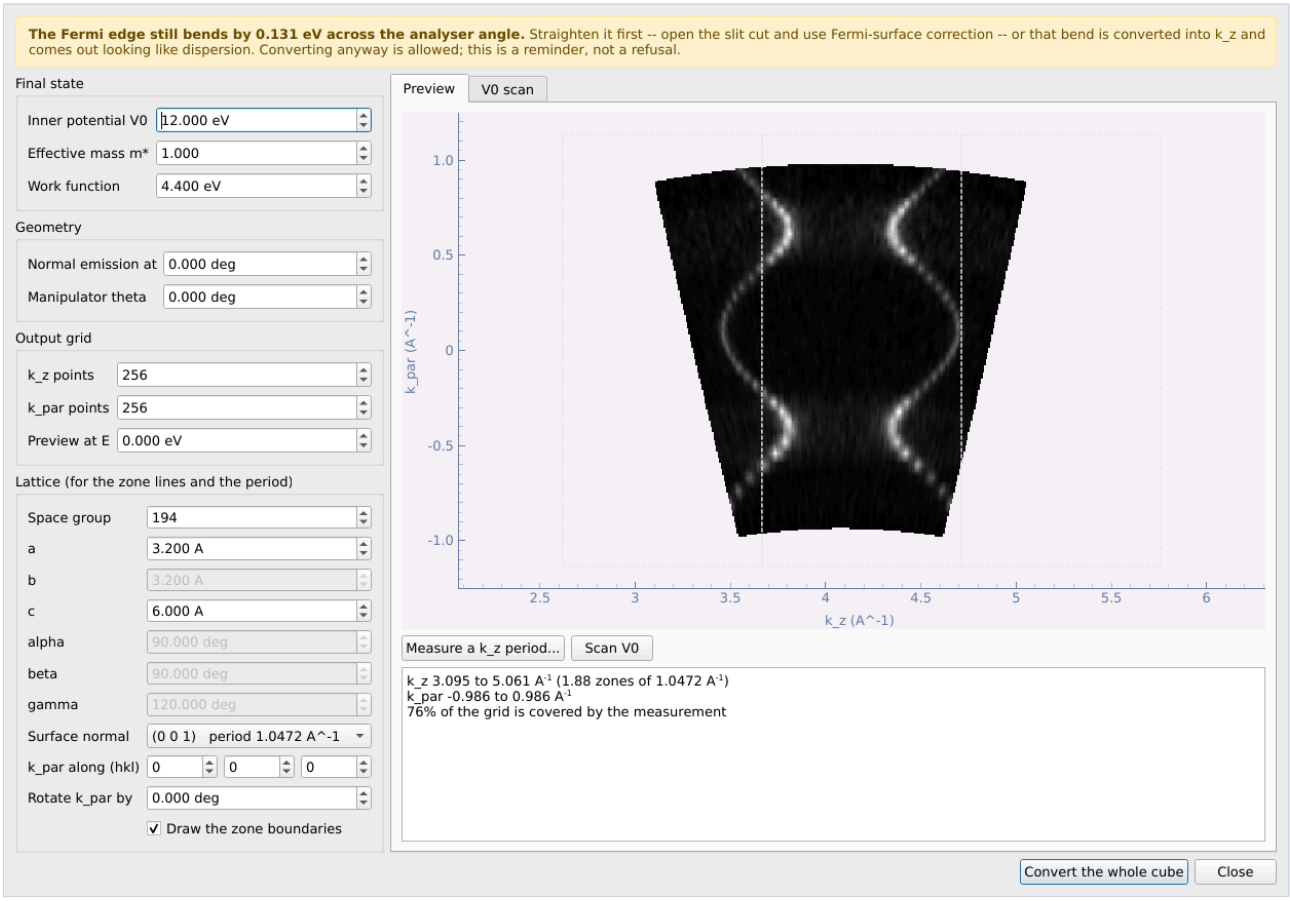
2. **Fit and assemble**: the same box (by detector index) is used for every spectrum; each is shifted to put its own edge at zero; the cube is cropped to the range they all cover; and, by default, each spectrum is **normalised by its total intensity**, which divides out the photon flux.



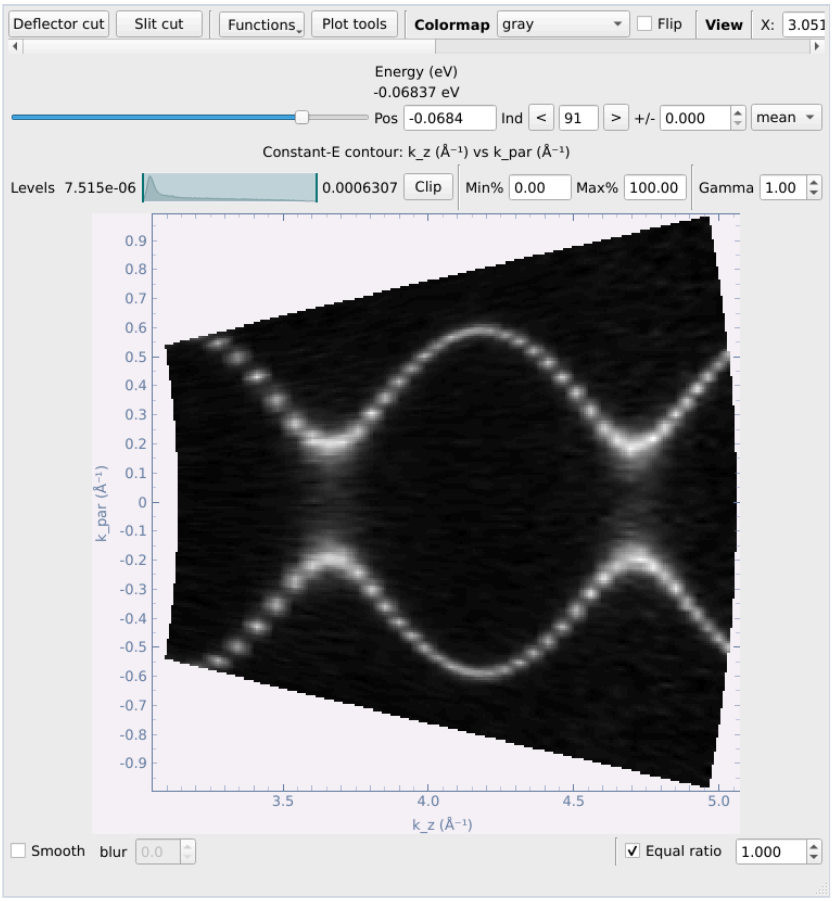
E_F against photon energy is plotted. A smooth curve means the fits are sound; points whose fit is not believed are marked and filled in from their neighbours.

13.2 Converting a kz map to momentum

Functions → Convert to momentum (kz, k∥)... converts the scan to (k_z, k∥).



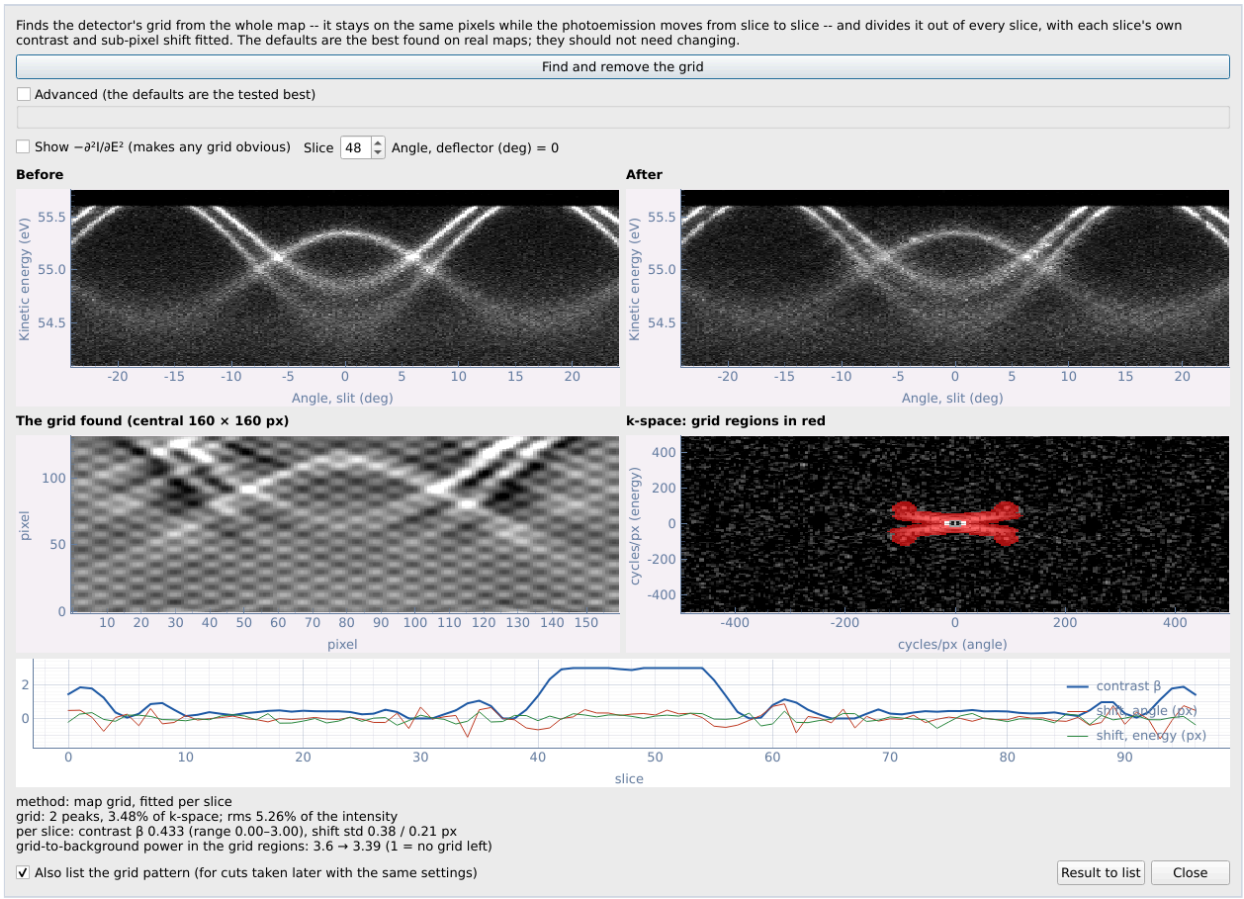
- **Inner potential V0** is the unknown: the preview reconverts one energy slice as you change it, in milliseconds.
- With the **lattice** (space group, a, b, c), the **surface normal** and the in-plane direction, the zone boundaries are drawn over the preview.
- **Scan V0** converts at a range of V0 values and reports where the k_z period matches the lattice — and how well that pins V0.
- **Measure a k_z period...**: click two points that are the same feature one zone apart; the matching reciprocal-lattice planes are listed.
- The yellow banner warns when the Fermi edge still bends across the analyser angle — correct it first (Straighten bands (FS correction) on the slit cut).
- **Convert the whole cube** lists a `kz map (k)`.



14. Removing the detector grid

MCP (hexagonal) and mesh (square) patterns print on every image. **Do it first**: only data still on the detector's pixels can be de-gridded, so the button refuses anything already converted, corrected or smoothed, and says why.

Functions → Remove detector grid... on a map (or its slit cut), or on a cut. For a map, no reference is needed — the grid stays on the same pixels in every slice while the photoemission moves. **Find and remove the grid**:

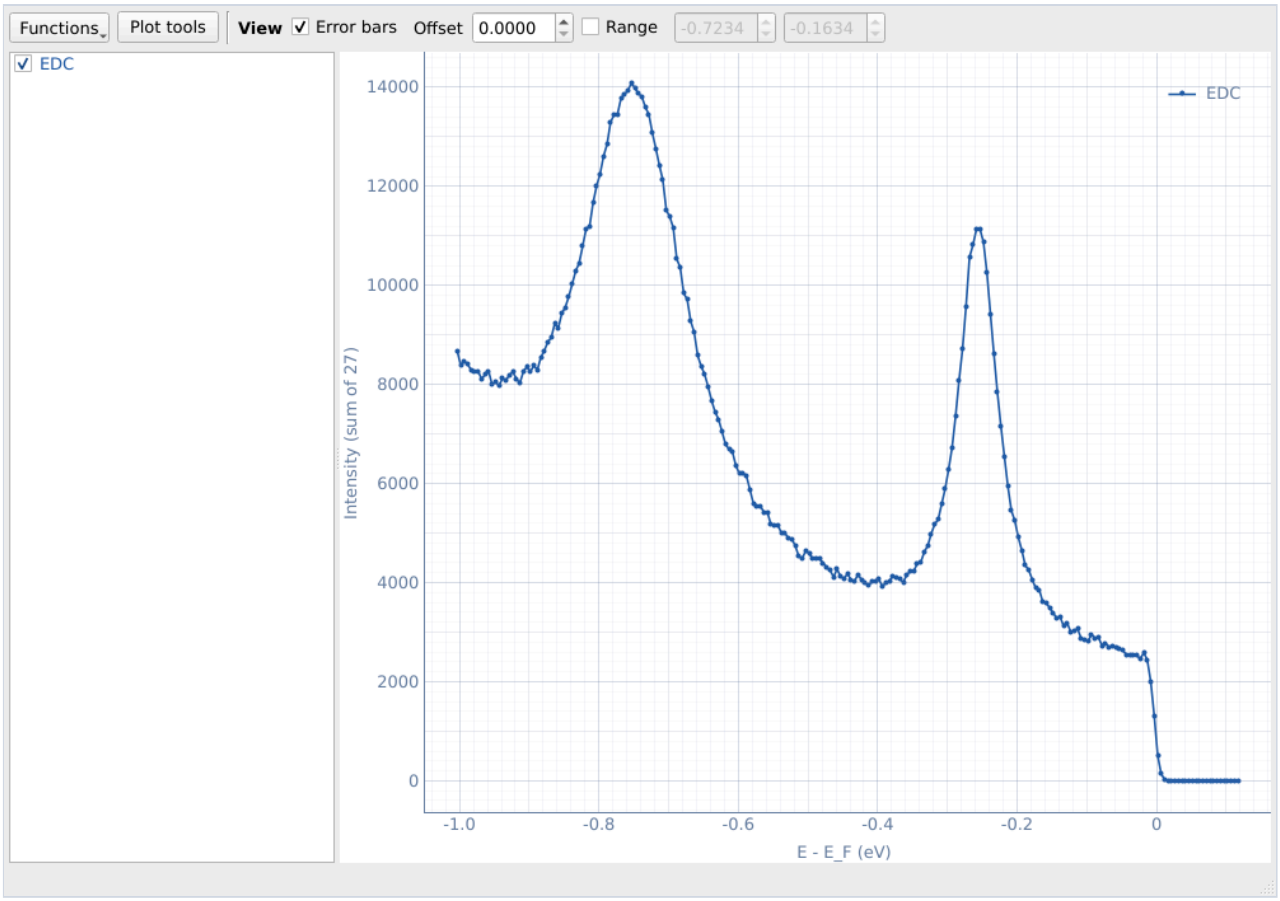


- **Before / After** a chosen slice on one colour scale (tick **Show −∂²I/∂E²** to make any grid obvious).
- **The grid found**, its **k-space regions** (in red), and the **contrast and shift** fitted for every slice.
- The power left in the grid regions (1 = no grid left).
- **Also list the grid pattern** keeps a `[grid]` dataset, which a cut taken later with the same lens mode and pass energy will find and use.

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15. Curves: EDCs, MDCs and spin-resolved data

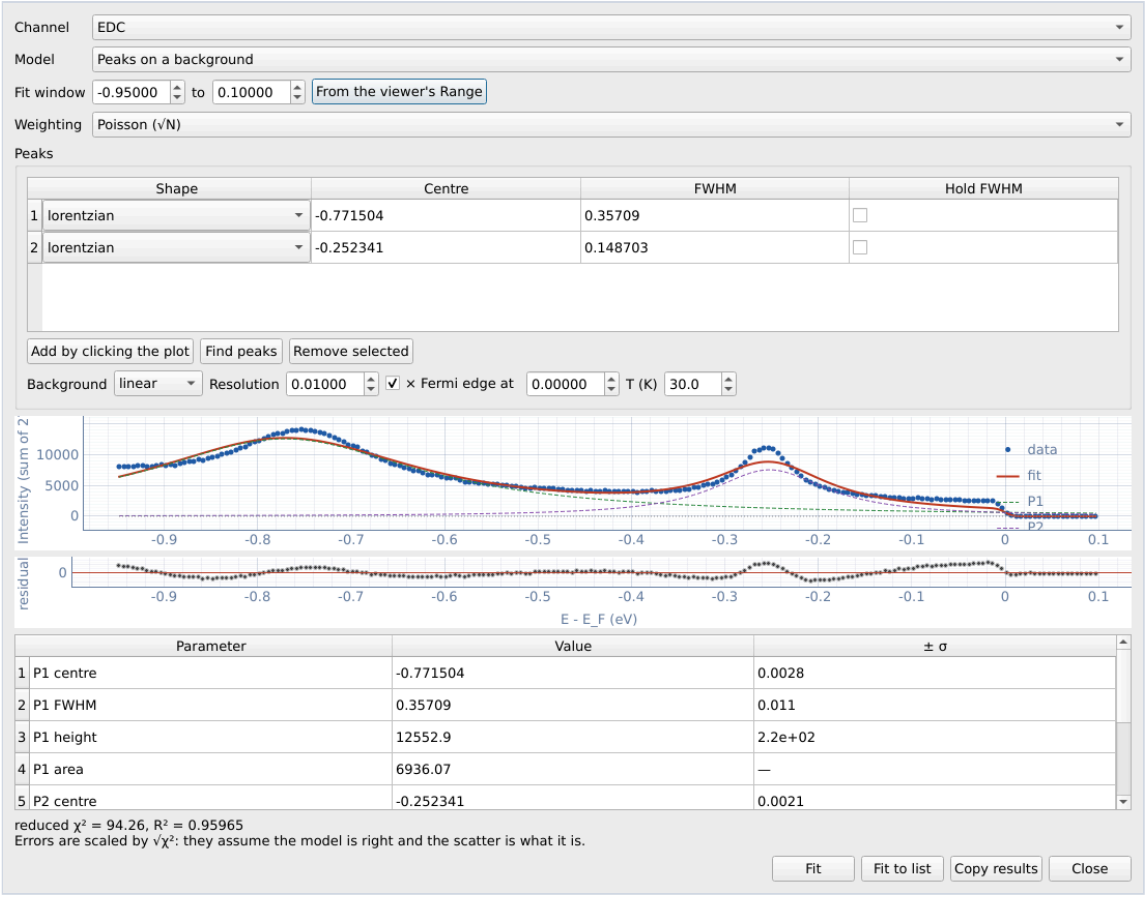
EDC → list / **MDC → list** under any image, or a spin-resolved file, give a curve dataset, which opens in the **curve viewer**:



Channels can be hidden, offset into a waterfall (**Offset**), shown with **Error bars** (from a σ channel), and **Range** marks an x window used by the operations. *Functions → Transform* gives **Crop**, **Bin**, **Normalise**, **Subtract a background** (constant, linear, Shirley), **Shift the axis** and **Add counting errors (√N)**; each makes a new dataset and propagates σ correctly.

15.1 Curve fit

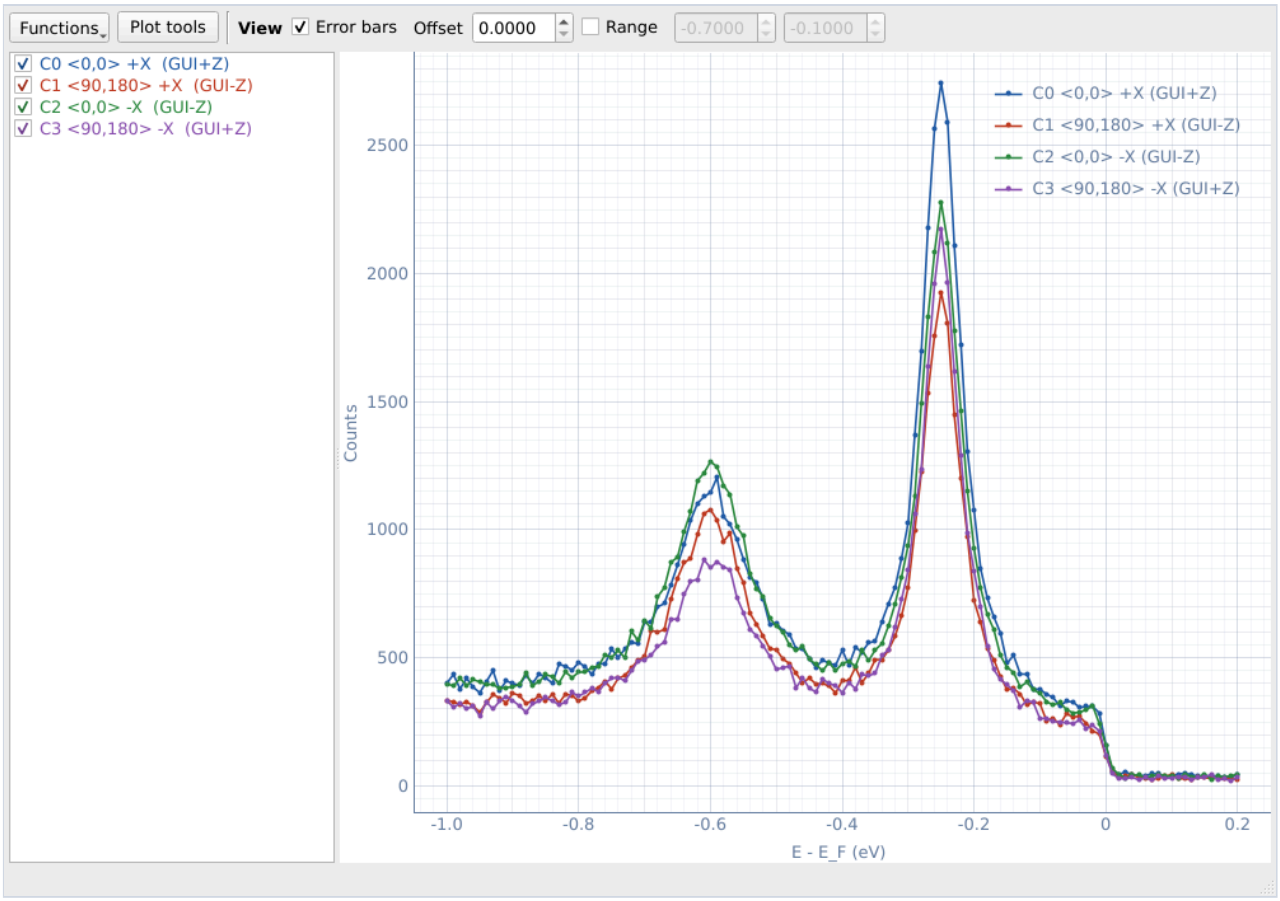
Functions → Curve fit…: **Peaks on a background** (Lorentzian, Gaussian or Voigt peaks placed by clicking or **Find peaks**; none / constant / linear / Shirley background; instrumental resolution; optional **× Fermi edge**), or the **Fermi edge** itself. Weighting is the channel's own σ, √N, or uniform. Every parameter comes with its error; the residual is shown in units of σ.



Fit to list keeps data, fit, residual and components; **Copy results** copies a tab-separated table. A Fermi-edge fit can **List a copy with E_F = 0**.

15.2 Spin analysis

A spin EDC opens with one curve per spin channel:



Functions → Spin analysis…:



- The table reads each channel's label (manipulator setting, target magnetisation, axis and sign counted as "up"); **Axis** and **Counts as** can be overridden.

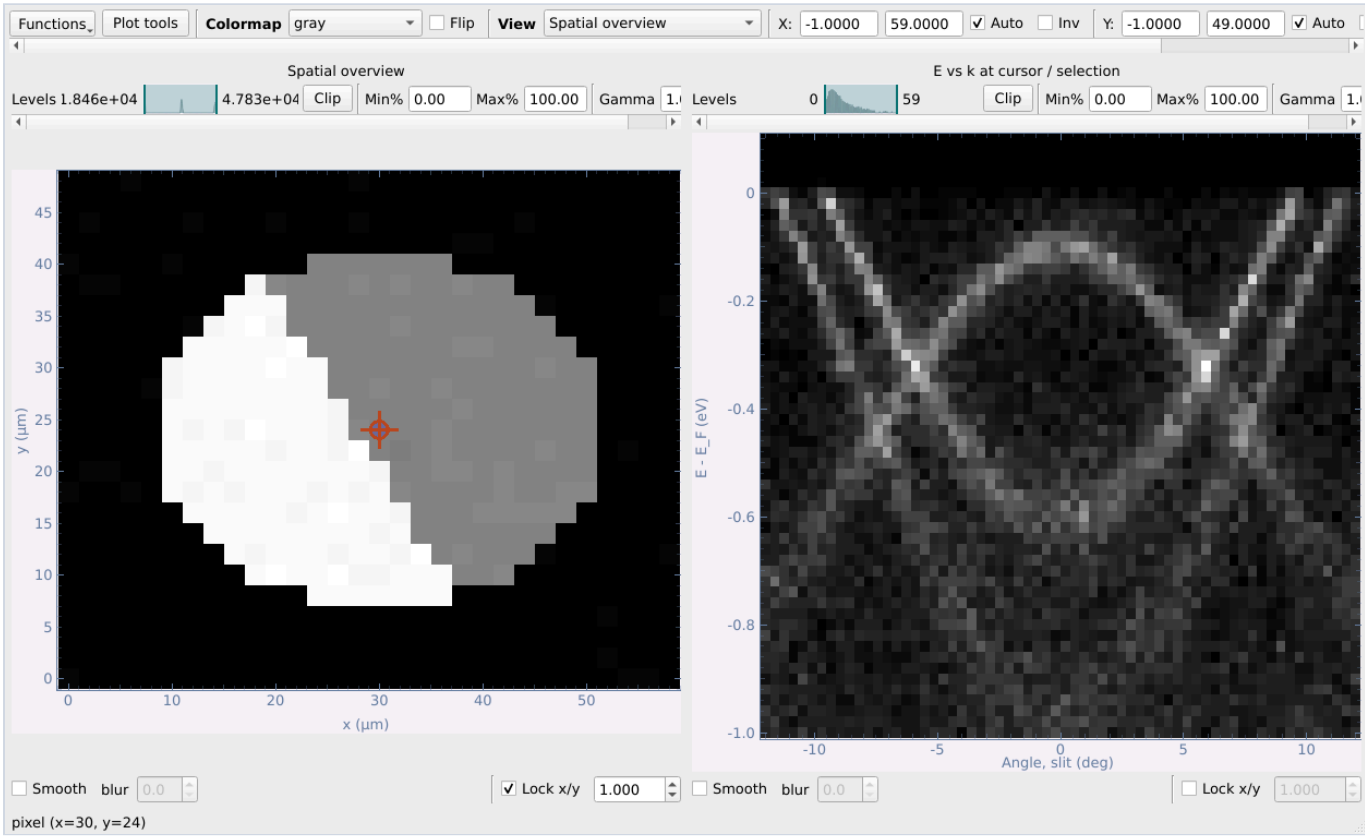
- **Cross ratio** (default when the channels allow it) cancels both the transmission and the target reflectivity exactly; **one pair** is available but carries the instrumental asymmetry, which is measured and reported (ε).
- **S_eff** (Sherman function, default 0.2), **Bin**, optional **Zero reference** window.
- Plots: raw channels, P with its error bars, and I↑ / I↓ with errors.
- **P to list**, **I↑ / I↓ to list**, **As a figure** (opens the figure composer).

---

16. Spatially resolved scans (nano-ARPES)

A real-space scan opens the **spatial map** (left, the total intensity at each position) together with the **E vs k spectrum at the cursor** (right). Drag the red target to read the spectrum at another point. Right-click a panel for the two integrations:

- on the map, **Integrate selection → E vs k panel** sums the spectra over a region of the sample;
- on the spectrum, **Integrate selection → spatial panel** maps the intensity of a chosen (k, E) window across the sample — how a domain or a flake edge is found.



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17. Figures and export

17.1 Figure composer

Select cuts (or curves) in the list → right-click → **Open in figure composer...** (or, from a frozen slice, its **Figure composer...** button). The composer lays panels out for a paper: one size per column, labels only on the outer edges, (a)/(b)/(c) letters, shared colour scales if you want them.

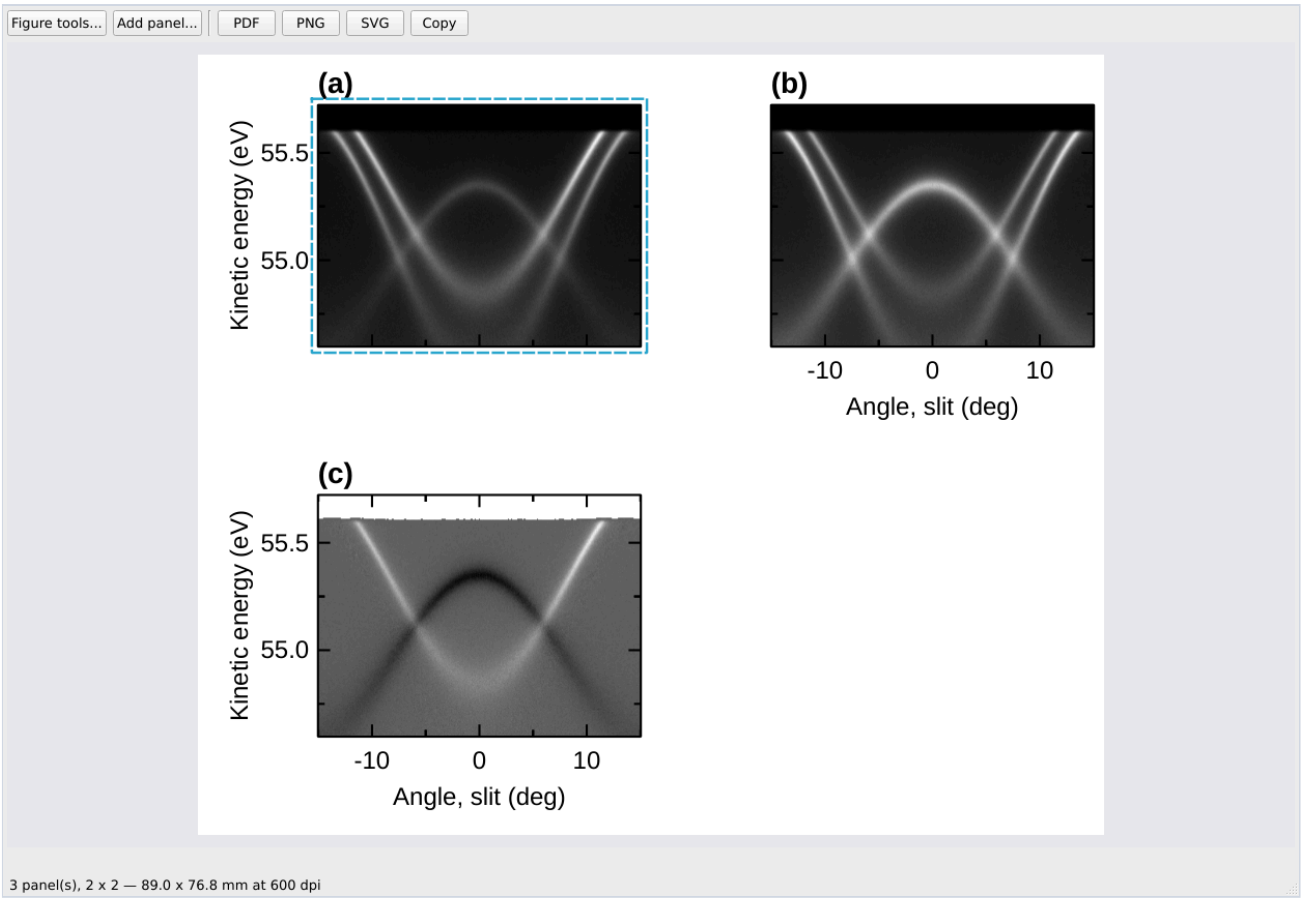
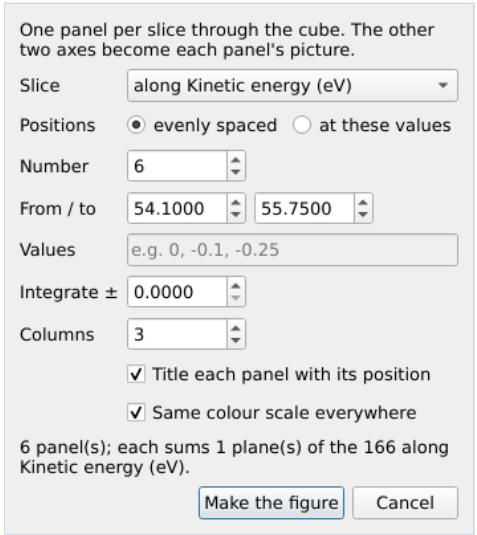
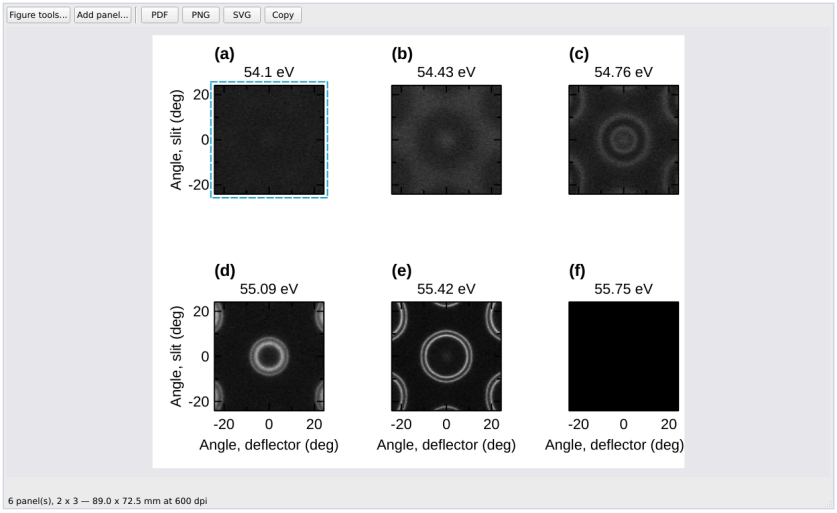


Figure tools... sets everything (size in mm, fonts, ticks, colour bars, lines, arrows, text, insets, scale bars) — not to be confused with a viewer's **Plot tools**, which draw guides on the live plot; **Add panel...** adds another dataset; **PDF**, **PNG**, **SVG** export, **Copy** puts it on the clipboard.

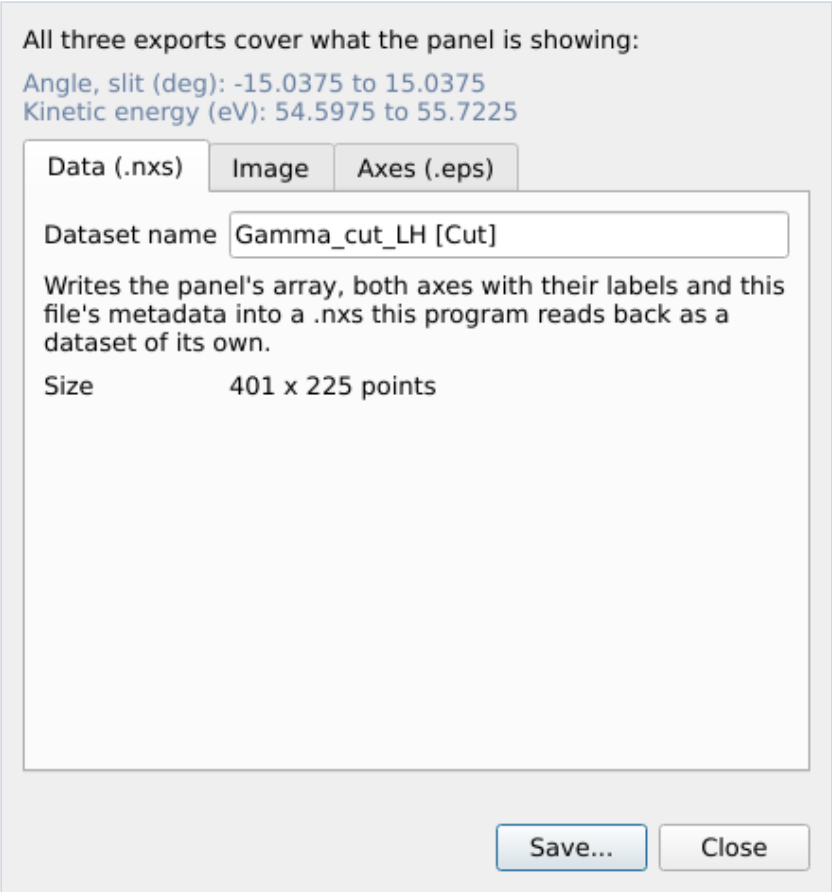
17.2 Multi-slice figure

From a map: *Functions → Multi-slice figure...* lays out a series of slices through the cube — constant-energy contours, or cuts — evenly spaced or at chosen values, with integration, one shared colour scale and titles.

Settings	Result
	

17.3 Exporting a panel

Right-click any image → **Export** → **Data (.nxs)...**, **Image...** or **Axes only (.eps)...**. It always exports exactly what the panel shows (range and all): the data as a dataset this program reads back; an image with the colormap applied (PNG, TIFF or JPEG, at a chosen size and dpi, with or without the axes, and with the Plot tools guides if there are any); or the frame, ticks and labels as vector PostScript to place the image in, in Illustrator.



Save (right-click rows in the list) writes datasets to a `.nxs` file of your own; several selected rows go into one file.

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18. Autosave, memory and recovery

- **Autosave.** Every computed dataset (a k-map, a processed cut, a fit result…) is written, uncompressed, to this session's folder in `~/.arpes_viewer/sessions/` the moment it exists. Real-space scans are not (they are the measurement itself, and large).
- **A normal close deletes the autosaved copies.** When you close the main window and confirm (everything you want is saved, or *Close anyway*), the open viewers are closed and the session folder is deleted straight away. A dataset's copy also goes as soon as you save it to a file of your own.
- **Recovery after a crash.** Only if the program did not close normally (a crash, a kill) is the session folder kept: the next start offers its datasets back, and it is deleted after three days.
- **Operations log.** `~/.arpes_viewer/sessions/operations.log`, one line per store / save / discard — how to find what was where.

- **Memory box.** This program's memory, the system's, how many viewers and HDF5 files are open, and how much computed data is held. **Free memory** drops cached copies of computed data (they are re-read from their autosaved file when needed); **Close all viewers** closes every window and then frees memory.
- **Open auto-save folder** (below the Memory box) shows the session folders in the system's file browser, to delete old working copies by hand. The folder of the session now running (named in the status bar) is in use.
- **Long operations** (reading a big map, converting, de-gridding, assembling a folder) run off the GUI thread with a progress bar and a working Cancel.
- **Closing** with unsaved computed data asks first, and names the datasets: **Save to a file...**, **Close anyway** (the autosaved copies are deleted) or **Cancel** (nothing happens).

19. Tips and troubleshooting

- **"Not available here" in the Functions menu** — hover over the entry: the tooltip says why (e.g. the MDC/EDC fit needs a cut already in $Å^{-1}$).
- **A k conversion is not offered** — the first axis of the map is not an angle (a photon energy, temperature…). Check *First axis of a map is* in the loader.
- **Cut k conversion of an E − E_F cut** — the energy offset is filled in for you from the Fermi fit; if you shifted the axis some other way, type the kinetic energy of E_F (photon energy − work function) into *Energy offset*.
- **A file is misread** — choose the reader explicitly in the loader, and see *Assumptions worth knowing about* in the README (axis order in SOLEIL files).
- **Dragging the cursor on a big map is sluggish** — a map read lazily from its file re-slices in about 0.1 s per window, so the other windows wait for the mouse to pause. Once the map has been read into memory it follows live.
- **Typing numbers** — every number box takes a decimal point (`0.05`), whatever the system language; a comma is not accepted.
- **The machine is paging** — watch the Memory bar (amber past a third of the RAM, red past a half); use **Free memory** or **Close all viewers**.